# COMPETITION BETWEEN RADIAL DIFFUSION AND INTERFACIAL EXCHANGE IN THERMAL-GRADIENT DEPOSITION COLUMNS: CONSERVATIVE MODELLING OF IODINE TRANSPORT

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## ABSTRACT

We present an OpenFOAM finite-volume model of vapour transport in laminar thermal-gradient columns: nine Pb–Bi–I gases in a frozen helium carrier, with element-conserving gas equilibrium and shared wall inventories. Reversible iodide exchange follows a Hertz–Knudsen–Schrage law in series with the near-wall diffusive resistance; on a Graetz–Robin problem this face flux converges at second order, a cell-centre sink at first order. In 79 simulations the largest raw element-balance residue is $1.41 \times 10^{-12}$ of supplied material. Halving the time step or adding correctors changes iodine profiles by at most 0.323%. First-order upwind advection differs from a bounded limited-linear scheme by up to 20.8% at equal mesh. Refining from 41,472 to 93,312 cells changes profiles by 2.13–7.30% (upwind, four cases) and 1.98% (bounded, one case), with inconsistent observed orders. A wall Damköhler sweep reproduces the kinetic and diffusion limits and shows that radial diffusion, not interfacial kinetics, limits wall exchange in the production cases; with the bounded scheme, accommodation changes alter iodine profiles by at most 5.14%, diffusivity changes by up to 16.65%. A $+29\,\mathrm{kJ\,mol^{-1}}$ shift in the BiI gas Gibbs energy changes the $BiI_3$ share of deposited iodide iodine from 94.58% to 2.01%. Without tuning, the wall law gives deposition-onset temperatures of five $PbI_2$ experiments with an RMS deviation of 18 K, against 35 K for two published models; because carrier, vapour-pressure and source assumptions differ and computed peaks lie 1–4 cm upstream, this promising result does not yet isolate the wall treatment. The results support a conservative method, not independent predictive validation.

***Keywords*** Thermal-gradient tube flow · Vapour deposition · Hertz–Knudsen–Schrage · Laminar mass transfer · Wall Damköhler number · Finite-volume method · Numerical verification

## Nomenclature

**Latin symbols**

| | |
|---|---|
| $A_f$ | wall face area, $\mathrm{m}^2$ |
| $a_{\mathrm{Pb}}$, $a_{\mathrm{Bi}}$ | activities of Pb and Bi in LBE |
| $b_k$ | concentration of element $k$, $\mathrm{mol\,m^{-3}}$ |
| $C_e$, $s_i$ | multipliers of the HKS conductance |
| $c$, $\bar{c}$, $c_{\mathrm{eq}}$ | vapour concentration, its cross-section mean and its equilibrium value (channel), $\mathrm{mol\,m^{-3}}$ |
| $D_i$ | molecular diffusivity of gas $i$, $\mathrm{m^2\,s^{-1}}$ |
| $d$ | tube diameter, m |
| $d_f$ | distance from cell centre to wall face, m |
| $G_i$ | HKS conductance, $\mathrm{s\,m^{-1}}$ |
| $G_{i,f}^{\mathrm{eff}}$ | effective (kinetic and half-cell diffusion) conductance, $\mathrm{s\,m^{-1}}$ |
| $G^\circ$ | standard Gibbs energy, $\mathrm{J\,mol^{-1}}$ |

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| | |
|---|---|
| $h$ | channel half-height, m |
| $J$ | wall molar flux in the channel, mol m$^{-2}$ s$^{-1}$ |
| $j_i$ | wall mass flux, positive from condensate to gas, kg m$^{-2}$ s$^{-1}$ |
| $K_{f,i}$ | formation equilibrium constant of gas $i$ |
| $K_r$ | equilibrium constant of the BiI reaction |
| $k_w$ | first-order wall uptake velocity, m s$^{-1}$ |
| $M_i$ | molar mass, kg mol$^{-1}$ |
| $m_i$ | source mass of compound $i$, kg |
| $\dot{m}$ | helium mass flow, kg s$^{-1}$ |
| $N_k$ | molar inventory of element $k$, mol |
| $N_r$ | number of radial cells |
| $p$ | pressure, Pa |
| $p_i$ | partial pressure of gas $i$, Pa |
| $p_{\mathrm{eq},i}$ | equilibrium pressure over the condensate, Pa |
| $p^\circ$ | standard pressure, $10^5$ Pa |
| $p_h$ | source pressure, Pa |
| $q_{1,i}$, $q_{2,k}$ | source rates of models Q1 and Q2, mol s$^{-1}$ |
| $q_{\mathrm{He}}$ | helium molar flow, mol s$^{-1}$ |
| $R$ | universal gas constant, J mol$^{-1}$ K$^{-1}$ |
| $R_k$ | balance residue of element $k$, mol |
| $R_t$ | tube radius, m |
| $r$, $x$ | radial and axial coordinates, m |
| $S_i^{\mathrm{rel}}$, $S_i^{\mathrm{wall}}$ | release and wall sources, kg m$^{-3}$ s$^{-1}$ |
| $T$ | temperature, K |
| $T_h$ | source-furnace temperature, K |
| $t$ | time, s |
| $U_b$ | bulk velocity, m s$^{-1}$ |
| $\mathbf{u}$ | carrier velocity, m s$^{-1}$ |
| $Y_i$ | mass of gas $i$ per unit carrier mass |
| **Greek symbols** | |
| $\alpha_i$ | accommodation coefficient |
| $\beta$ | metal saturation of source model Q2 |
| $\kappa$ | axial decay rate of the Graetz problem, m$^{-1}$ |
| $\lambda_k$ | dimensionless element potential |
| $\mu$ | dynamic viscosity, Pa s |
| $\nu_{ki}$ | atoms of element $k$ in gas $i$ |
| $\rho$ | carrier density, kg m$^{-3}$ |
| $\tau_{\mathrm{exp}}$ | duration of the experiment, s |
| $\chi_i$ | $\rho RT/M_i$, Pa |
| **Dimensionless numbers** | |
| Bi | Biot number, $k_w R_t/D$ |
| $\mathrm{Da}_i$ | wall Damköhler number, eq. (14) |
| Gr | Grashof number |
| $\hat{J}$ | normalised wall flux, $Jh/[D(\bar{c}-c_{\mathrm{eq}})]$ |
| Pe | Péclet number, $\mathrm{Re}\,\mathrm{Sc}$ (diameter based) |
| $\mathrm{Pe}_R$ | radius-based Péclet number of the Graetz problem, $U_b R_t/D$ |
| Re | Reynolds number, $4\dot{m}/(\pi d\mu)$ |
| Sc | Schmidt number |
| **Subscripts** | |
| $c$ | cell centre |
| $f$ | wall face |
| $i$ | gas species |
| $k$ | element (Pb, Bi, I) |
| $w$ | wall |

# 1 Introduction

Iodine released from lead–bismuth eutectic (LBE) can travel in several chemical forms before it deposits on cooler surfaces. The chemical form matters because it sets both the gas-phase inventory available for transport and the equilibrium pressure that governs wall exchange. Liu et al. [1] studied this behaviour by evaporating pure iodides and iodine-containing LBE into helium, separating the products along a negative temperature gradient and characterising the deposits. Their measurements give spatially resolved evidence for $PbI_2$ and $BiI_3$ deposition, and for potassium-containing deposits in some LBE experiments. Linking such measurements to a transport calculation needs more than

a single deposition temperature: the temperature-coordinate mapping, the carrier flow, the source history and the gas composition all affect the predicted distribution.

Seen as a fluid-dynamics problem, a thermal-gradient column is a slow laminar flow of an inert carrier through a long, narrow tube whose wall temperature falls by several hundred kelvin along its length (fig. 1). Trace vapours released near the hot end are advected downstream by a carrier whose density and velocity vary with the imposed temperature, and they diffuse radially towards the wall. Where the local vapour pressure exceeds the equilibrium pressure at the wall temperature, material is lost to the wall; where it falls below that pressure, existing deposits can evaporate again. The deposition profile is therefore set by the competition between axial advection, radial diffusion to the wall and the interfacial exchange law. This is a Graetz-type wall mass-transfer problem [2], but with a nonlinear wall condition that depends on the local wall temperature and condensed inventory. The same coupling underlies the interpretation of thermochromatographic experiments [3–5].

Existing CFD treatments of such wall exchange differ mainly in where the exchange is applied and in how many coupled species they carry. Lobresco et al. [6] compared a calibrated temperature-based deposition law with a vapour-pressure-driven Hertz–Knudsen–Schrage law for $PbI_2$, both formulated as volumetric source terms rather than boundary fluxes. For steam condensation with non-condensable gases, Vijaya Kumar et al. [7] implemented a diffusion-layer wall model in the OpenFOAM-based containmentFOAM both as volumetric sources and as face fluxes, and found the face-flux form more accurate, cheaper and less sensitive to the grid. Mixed (Robin) deposition boundary conditions for fission-product transport in OpenFOAM have been verified against analytical solutions [8]. In these studies each depositing or condensing species is transported on its own; gas-phase changes of chemical form are not coupled to the wall exchange.

This paper adds that coupling, in the finite-volume framework of OpenFOAM [9]. Our contribution has three parts. First, element-conserving homogeneous equilibrium among nine Pb–Bi–I gases is coupled to reversible face-flux wall exchange, in which the interfacial kinetic conductance acts in series with diffusion over the distance from the cell centre to the wall; the Graetz–Robin verification shows that this face flux is second-order accurate where the same kinetic law applied to the cell-centre value, the form of a first-cell sink, is first order. Second, gas pathways that lead to the same condensed material share one physical wall inventory, so that switching between deposition, evaporation and reaction cannot create or duplicate material. Third, we characterise the transport behaviour that results in these columns: wall exchange limited by radial diffusion rather than by interfacial kinetics, which a sweep of the wall Damköhler number over six decades confirms by reproducing both limits of the exchange and the transition between them; profiles whose local shape is far more resolution-sensitive than their integrals; and an iodide partition that is controlled by thermodynamic input. The software also has an interface to the GEMS3K kernel of GEM-Selektor [10], but the reported calculations use a specialised ideal-gas element-potential solver. Repeating one production case with GEMS3K standard-state data in every cell changes its deposit profiles by at most $2.1 \times 10^{-4}$ (relative $L_1$); a mode that also calls the GEMS3K solver in every cell ($5.1 \times 10^8$ calls, none failed) gives bitwise identical results, because that solution only initialises the element-potential solve.

All experimental input comes from published papers, so we report the consequences of two explicit source models and of selected input variations rather than infer a unique physical source. The study addresses three questions. Can transport, gas equilibrium and competing wall channels be coupled while preserving each represented element? How strongly do deposition profiles depend on time step, iteration, mesh and advection scheme, including when total deposition appears stable? Which input changes materially alter the interpretation of the computed deposits? The evidence comprises 79 computational cases of the three available Liu et al. configurations (not experimental replicates), production re-runs with a bounded advection scheme, a scheme and mesh study, an iteration check, the Graetz–Robin verification, and an untuned comparison with five single-species experiments of Lobresco et al. and with the two models of that study. The scope is the executed model: a fixed carrier and temperature field, prescribed release, ideal-gas speciation, and wall deposits without aerosol formation or substrate chemistry (section 8.3). Comparisons with experiment are conditional assessments of this model, not independent predictive validation.

Sections 2 to 4 define the reconstructed inputs, equations and case matrix. Section 5 verifies the wall discretisation and assesses conservation and resolution, and sections 6 and 7 examine experimental discrepancies and input sensitivity. Section 8 states what the evidence supports and collects the limitations.

## 2 Experimental basis and input reconstruction

### 2.1 Available configurations

Liu et al. [1] used 1.25 m columns with an inner diameter of 4.8 mm for 316L stainless steel and 5.0 mm for fused silica. A 15 mm sample boat was inserted into a hot starting furnace, followed by a second furnace that imposed

Table 1: Published configurations used in the baseline matrix. Sample masses are initial masses, not the amounts released in a 60 s run. Every experimental evaporation period was 10,800 s.

| Identifier | Column | Diameter (mm) | Sample (mg) | He (mL/min) |
|---|---|---|---|---|
| PbI2_SS | Steel | 4.8 | 14.6 | 100 |
| LBE-I_SS_II | Steel | 4.8 | 346.0 | 45 |
| LBE-I_SiO2_I | Silica | 5.0 | 368.3 | 45 |

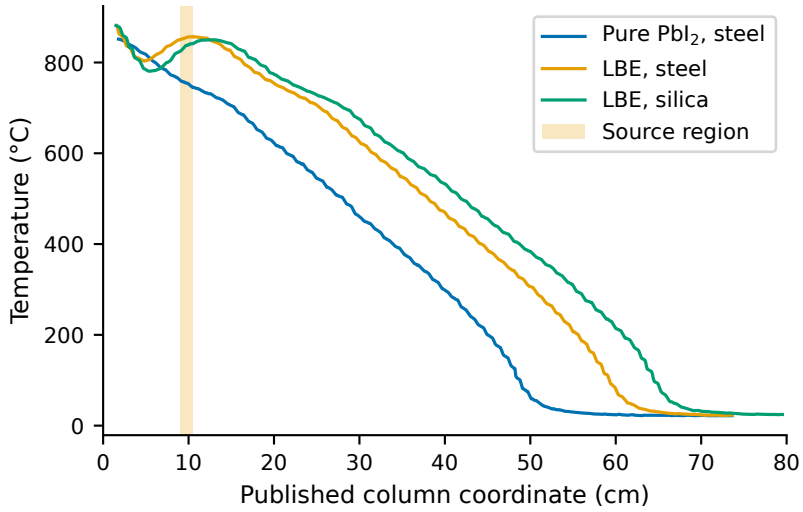



Figure 1: Axial temperature curves digitised from Liu et al. [1] for the three configurations, with the assumed computational source interval. Beyond the plotted range, the CFD domain holds the endpoint values constant. The curves reconstruct published figures; they are not temperature fields predicted by a conjugate heat-transfer calculation.

a decreasing temperature profile. Helium carried the evaporated material downstream. After three hours the source was withdrawn, the furnaces and gas flow were turned off, and the column cooled before the deposits were analysed. Temperature measurements and radiotracer scans used 1 cm increments.

Table 1 lists the three configurations for which both temperature curves and iodine scans could be reconstructed. The stand-alone $BiI_3$ experiment at 800 °C was not run, because its axial temperature profile is not published; it is neither given a surrogate profile nor counted as an unsuccessful simulation.

Both LBE samples had a reported iodine mole fraction of $9.04 \times 10^{-4}$ and a 900 °C starting furnace. Their reported deposited iodine amounts were 0.191 and 0.203 mg, respectively. These amounts set the prescribed mean iodine source in the present cases; they are not measurements of a time-resolved evaporation rate or of inlet speciation.

### 2.2 Digitisation and coordinate assumptions

We digitised the axial temperature curves and iodine histograms from the published figures. Temperatures are interpolated linearly to cell centres and wall faces; beyond the plotted interval they are held at the nearest endpoint value. The computational coordinate is the plotted coordinate, whose origin relative to the physical boat and furnace is unconfirmed (fig. 1); the silica-coordinate study varies this assumption.

The recorded digitisation resolutions are approximately 1 mm in position, 4 K in temperature and 0.4 percentage points in histogram height. They describe raster extraction only, not coordinate registration, detector response, temperature reproducibility or calibration, and the uncertainties printed with the deposition temperatures are not uncertainties of every profile bin. The reconstructed histograms sum to 99.17%, 96.70% and 102.24% for pure $PbI_2$, steel LBE and silica LBE. The primary comparison keeps these sums and does not silently renormalise them.

In the LBE scans, the paper assigns 6.6% and 7.0% of the iodine to KI, which lies outside the Pb–Bi–I species set. We scale the model profile to the remaining 93.4% or 93.0%, while the measured histogram keeps all its published deposits. A second, explicitly conditional comparison normalises both profiles over their common scanned range (section 6). Neither comparison supplies the omitted potassium chemistry.

### 2.3 Simulation duration and measurement protocol

All source-driven cases run for 60 s with a source rate based on the 10,800 s experimental period, so they release $1/180$ of the assumed three-hour inventory; the cumulative wall inventories inform deposition rates and conditional profile shapes but still contain the initial filling transient. Four additional cases stop the source and the carrier flow at 60 s and continue to 80 s with the original temperature profile. They probe the relaxation of residual gas, not the experimental cooling history (supplementary material, Sec. S1).

## 3 Transport, speciation and wall exchange

### 3.1 Frozen carrier and trace-species equations

The carrier fields $\rho$, $\mathbf{u}$, $T$ and $p$ come from a steady laminar helium calculation and are then held fixed. The reconstructed axial temperature is imposed over each whole cross-section. A gas species $i$ is represented by $Y_i$, its mass per unit carrier mass, and satisfies

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + S_i^{\mathrm{rel}} + S_i^{\mathrm{wall}}. \tag{1}$$

Gas equilibrium is applied as a local redistribution between transport steps. Species do not alter the carrier density, momentum or temperature. Here $D_i$ is the molecular diffusivity and $S_i^{\mathrm{rel}}$ and $S_i^{\mathrm{wall}}$ are volumetric mass sources ($\mathrm{kg\,m^{-3}\,s^{-1}}$); the latter is positive for net wall release. Because $Y_i$ is defined relative to carrier mass, the nine reactive-species variables are not constrained to sum to unity. The represented gases are Pb, PbI, $PbI_2$, Bi, $Bi_2$, BiI, $BiI_3$, I and $I_2$. Helium is inert. Condensed $PbI_2$, $BiI_3$, Pb and Bi are stored as immobile wall inventories. The solid-state identifiers in the code label these reservoirs; the equilibrium-pressure tables select between crystalline and liquid states where applicable.

The model does not represent bulk nucleation, aerosols and their thermophoretic deposition [11, 12], thermal (Soret) diffusion [13, 14], hydrogen/oxygen chemistry, latent-heat feedback or substrate reactions, so differences between the columns come from their thermal profiles, dimensions and sources, not from a substrate-specific sticking law.

### 3.2 Homogeneous ideal-gas equilibrium

Let $\nu_{ki}$ be the number of atoms of element $k \in \{\mathrm{Pb}, \mathrm{Bi}, \mathrm{I}\}$ in gas $i$. At fixed local temperature the transported element concentrations are

$$b_k = \sum_i \nu_{ki} \frac{\rho Y_i}{M_i}. \tag{2}$$

The equilibrium solver finds the gas partial pressures and dimensionless element potentials $\lambda_k$ that satisfy

$$\begin{aligned} \frac{p_i}{p^\circ} &= K_{f,i}(T) \exp\left( \sum_k \nu_{ki} \lambda_k \right), \\ \sum_i \nu_{ki} \frac{p_i}{RT} &= b_k, \qquad p^\circ = 10^5\ \mathrm{Pa}. \end{aligned} \tag{3}$$

The tabulated $K_{f,i}$ share a common elemental reference and the stated standard pressure; the thermodynamic records, their provenance and the phase treatment are summarised in Sec. S5 of the supplementary material. Concentrations are converted back through $Y_i = M_i p_i / (\rho RT)$. Re-speciation therefore conserves elements, not individual species. The helium pressure is not an additional reactive element constraint; under the trace-species approximation its field stays prescribed.

The solver works in logarithmic variables with a safeguarded Newton iteration. The Pb and Bi subproblems are reduced at fixed iodine potential, and a Schur complement treats the remaining coupling. Re-speciation is requested at every time step in cells with $T \geq 350\,\mathrm{K}$; below this threshold the transported gas composition is kept. This cutoff is a numerical modelling choice, not a demonstration that reactions cease at 350 K. Homogeneous condensed phases are excluded, and instantaneous gas equilibrium stands in for unmeasured finite reaction rates.

### 3.3 Reversible iodide exchange

For $PbI_2$ and $BiI_3$, the kinetic wall law defines the flux as positive from condensate to gas:

$$j_i = G_i \left[ p_{\mathrm{eq},i}(T_w) - p_{i,w} \right], \qquad G_i = C_e s_i \frac{2\alpha_i}{2 - \alpha_i} \sqrt{\frac{M_i}{2\pi R T_w}}. \tag{4}$$

Here $j_i$ is in $\mathrm{kg\,m^{-2}\,s^{-1}}$, $M_i$ is in $\mathrm{kg\,mol^{-1}}$, pressures are in Pa, $\alpha_i$ is the accommodation coefficient and $C_e$, $s_i$ are dimensionless multipliers. The baseline uses $C_e = s_i = \alpha_i = 1$. This is the Hertz–Knudsen–Schrage (HKS) form [15–17] used in vapour-pressure-driven deposition modelling [6]. The condensed-state pressure follows from the gas and condensed Gibbs energies on the stable tabulated condensed branch. The wall temperature selects this pressure; no deposition threshold is prescribed.

The kinetic law must also connect to the cell-centre concentration through molecular diffusion. With $d_f$ the cell-centre-to-wall distance, $p_{i,c} = \chi_{i,c} Y_{i,c}$ and $\chi_{i,c} = \rho_c R T_c / M_i$, eliminating the wall concentration gives

$$G_{i,f}^{\mathrm{eff}} = \frac{G_i}{1 + G_i \chi_{i,c} d_f / (\rho D_i)_f}, \qquad j_{i,f} = G_{i,f}^{\mathrm{eff}} [p_{\mathrm{eq},i}(T_w) - \chi_{i,c} Y_{i,c}]. \tag{5}$$

This combines kinetic and half-cell diffusion resistances in series. In particular, $1/G_{i,f}^{\mathrm{eff}} = 1/G_i + \chi_{i,c} d_f / (\rho D_i)_f$: the smaller conductance limits transfer. As $G_i$ increases, the effective conductance approaches the finite diffusive limit $(\rho D_i)_f / (\chi_{i,c} d_f)$. This separates the physical interfacial coefficient from the mesh-dependent distance used to connect the wall to the resolved gas field. The gas equation receives the face flow $A_f j_{i,f}$ as an implicit cell contribution, and the wall inventory receives its negative. Evaporation is limited by the face inventory available over the time step, so a bare wall cannot evaporate. Inventory updates use the exchange actually realised by the final gas solve, including active bounds, not an independently evaluated explicit predictor. Switching between deposition and evaporation therefore does not create material.

### 3.4 Shared metal inventories and a reactive channel

Additional wall channels connect Pb(g) to condensed Pb, and Bi(g) and $Bi_2$(g) to one common condensed Bi inventory. Each channel has its own equilibrium pressure and stoichiometric mass conversion, and all use the baseline interfacial coefficients. The reactive channel

$$3\mathrm{BiI}(g) \longrightarrow 2\mathrm{Bi(condensed)} + \mathrm{BiI_3}(g) \tag{6}$$

is enabled only in the forward direction in the executed cases. For unit activity of condensed Bi, the dimensionless reaction constant and effective reactant equilibrium pressure are

$$K_r(T_w) = \exp\left[ -\frac{2G^\circ_{\mathrm{Bi,c}} + G^\circ_{\mathrm{BiI_3,g}} - 3G^\circ_{\mathrm{BiI,g}}}{R T_w} \right], \tag{7}$$

$$p_{\mathrm{eq,BiI}} = p^\circ \left[ \frac{p_{\mathrm{BiI_3,c}}/p^\circ}{K_r(T_w)} \right]^{1/3}, \tag{8}$$

$$j_{\mathrm{BiI}} = \min\{0,\, G_{\mathrm{BiI}}^{\mathrm{eff}} [p_{\mathrm{eq,BiI}} - p_{\mathrm{BiI,c}}]\}. \tag{9}$$

All Gibbs energies in $K_r$ are evaluated at $T_w$. This phenomenological channel applies the HKS conductance, with the same half-cell resistance, to the reaction driving force; the coproduct pressure is taken from the beginning-of-step gas field, and the minimum suppresses the reverse reaction. The reaction returns one $BiI_3$ molecule to the gas and adds two Bi atoms to the shared wall inventory, on which all Bi channels draw; a separate inventory per reaction would duplicate a physical material. Coproducts are applied after the transported reactants have been solved. The channel is an irreversible model assumption, not a surface-equilibrium model.

### 3.5 Prescribed source models

Source model Q1 releases $PbI_2$ and, for LBE, $BiI_3$ at constant rates

$$q_{1,i} = \frac{m_i}{M_i \tau_{\mathrm{exp}}}, \qquad \tau_{\mathrm{exp}} = 10{,}800\,\mathrm{s}. \tag{10}$$

For pure $PbI_2$, $m_i = 14.6\,\mathrm{mg}$ is the initial sample mass, and complete uniform release over the experimental period is assumed. For steel LBE, $m_{\mathrm{PbI_2}} = 0.111\,\mathrm{mg}$ and $m_{\mathrm{BiI_3}} = 0.181\,\mathrm{mg}$; for silica LBE, they are 0.243 and 0.085 mg. These compound masses were inferred from the deposits reported in the experimental paper. Q1 therefore conditions both the iodine loading and much of the species split on the measurements.

Source model Q2 instead supplies atomic Pb, Bi and I to the gas-equilibrium system. With $q_{\mathrm{He}}$ the full-column helium molar flow and $p_h$ the source pressure, its elemental rates are

$$q_{2,\mathrm{Pb}} = q_{1,\mathrm{PbI_2}} + q_{\mathrm{He}} \beta p_{\mathrm{Pb}}(T_h)/p_h, \tag{11}$$

$$q_{2,\mathrm{Bi}} = q_{1,\mathrm{BiI_3}} + q_{\mathrm{He}} \beta [p_{\mathrm{Bi}}(T_h) + 2p_{\mathrm{Bi_2}}(T_h)]/p_h, \tag{12}$$

$$q_{2,\mathrm{I}} = 2q_{1,\mathrm{PbI_2}} + 3q_{1,\mathrm{BiI_3}}. \tag{13}$$

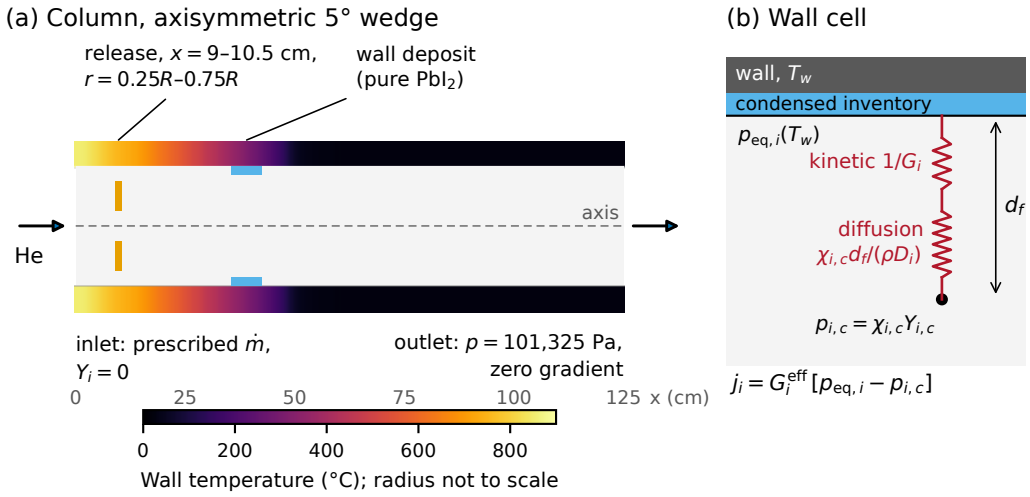



Figure 2: (a) Computational column, coloured by the digitised wall temperature of the pure $PbI_2$ steel experiment, with the prescribed release region, the boundary conditions of section 4.1 and the extent of the computed $PbI_2$ deposit (1 cm bins holding at least 1% of the peak, production case). The radius is not to scale, and the source is distributed over selected cells; no boat surface is resolved. (b) One wall cell: the HKS conductance (eq. (4)) acts in series with diffusion over the cell-centre-to-wall distance $d_f$ (eq. (5)); the equilibrium pressure is evaluated at the wall temperature.

The metal pressures use ideal LBE activities $a_{\mathrm{Pb}} = 0.447$ and $a_{\mathrm{Bi}} = 0.553$ at $T_h = 1173.15\,\mathrm{K}$. The multiplicative metal saturation $\beta$ is unmeasured. Q2 changes the metal supply and lets the iodide split emerge from the model, but its iodine source is still fixed by the deposited iodine. Q2 is therefore a source hypothesis, not an independent identification of LBE evaporation. The rates are distributed over an assumed source region, without resolving a liquid surface, depletion of the physical boat or the boat's obstruction of the flow.

# 4 Numerical implementation and study design

## 4.1 Geometry, carrier and boundary conditions

A structured axisymmetric 5° wedge represents the 1.25 m column (fig. 2). The tube radius $R_t$ is 2.4 mm for steel and 2.5 mm for silica. The production mesh has $576 \times 72 \times 1 = 41{,}472$ cells. The 64 follow-up sensitivities use the original $384 \times 48 \times 1 = 18{,}432$-cell mesh. The mesh studies use $256 \times 32 \times 1 = 8{,}192$, 18,432, 41,472 and $864 \times 108 \times 1 = 93{,}312$ cells, the finest for four cases with upwind advection and one with the bounded scheme; successive levels refine the axial and radial spacing by a factor of 1.5. The source occupies $x = 0.09$–$0.105$ m and the radial interval $r = 0.25R_t$–$0.75R_t$. Its position and cell selection are computational assumptions, and the physical boat is omitted from the carrier geometry. Full-column source and inlet rates are multiplied by $5/360 = 1/72$ for the wedge.

The inlet flow rates are converted to helium mass flow assuming a controller reference of 298.15 K and 101,325 Pa, which the supplied paper does not specify. Full-column rates are $2.727 \times 10^{-7}\,\mathrm{kg\,s^{-1}}$ at 100 mL/min and $1.227 \times 10^{-7}\,\mathrm{kg\,s^{-1}}$ at 45 mL/min; a separate study uses a 273.15 K reference. Helium follows the perfect-gas law with constant heat capacity and Sutherland viscosity (supplementary material, Sec. S5). Steady laminar carriers are computed with `rhoSimpleFoam` while the reconstructed temperature profile stays prescribed. The converged face mass flux is corrected to discrete continuity within $10^{-12}$ of the inlet flow (supplementary material, Sec. S2), which makes the frozen transport field conservative but is not evidence that the velocity field has converged.

The boundary conditions are the same in all Liu et al. cases. At the inlet the carrier enters with the prescribed mass flow at the local wall temperature, and every reactive gas has a fixed mass fraction of zero: clean helium enters, and the release is a volumetric source in the source cells. At the outlet the static pressure is fixed at 101,325 Pa, an assumed value because neither experimental paper states the column pressure, and all gases have zero normal gradient, so they leave by advection. At the wall the transported gases have zero normal gradient in the transport operator, and all wall exchange enters as the face fluxes of section 3.3. With the 101,325 Pa outlet pressure and the baseline diffusivity, the Schmidt number is 2.0–11.3 between 400 and 1173 K, and the diameter-based Péclet number $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Sc}$ is 5.9–17.4 at 100 mL/min and 2.6–7.8 at 45 mL/min. Axial diffusion is therefore not negligible next to advection, and the

fixed zero inlet value permits a diffusive loss of vapour through the inlet; it is included in the outward transport of the material ledger (eq. (15)). The flow-stop cases replace both end conditions by zero normal species flux (supplementary material, Sec. S1).

### 4.2 Transport regime

With the full-column helium mass flow $\dot{m}$, the tube diameter $d$ and the Sutherland viscosity, $\mathrm{Re} = 4\dot{m}/(\pi d\mu)$ is 1.5–3.0 for the 100 mL/min steel case and 0.7–1.3 for the 45 mL/min cases between the source-furnace temperature of 1173 K and 400 K; the carrier is laminar throughout. With the baseline diffusivity, the radial diffusion time $R_t^2/D_i$ is 58 ms in steel and 63 ms in silica. Placing the conductance of eq. (4) in series with diffusion over the tube radius, as in eq. (5) with $d_f$ replaced by $R_t$ and $T_c = T_w$, defines a wall Damköhler number

$$\begin{aligned} \mathrm{Da}_i &= \frac{G_i \chi_i R_t}{\rho D_i} \\ &= C_e s_i \frac{2\alpha_i}{2-\alpha_i} \sqrt{\frac{RT_w}{2\pi M_i}}\, \frac{R_t}{D_i}, \end{aligned} \tag{14}$$

where $R$ is the universal gas constant and $R_t$ is the tube radius. Because $\chi_i/\rho = RT_w/M_i$, it does not depend on pressure. For $PbI_2$ and $BiI_3$ between 400 and 1173 K, the baseline coefficients give $\mathrm{Da}_i \approx 1.4 \times 10^3$–$2.9 \times 10^3$, and the tested accommodation coefficients $\alpha_i = 0.1$ and 0.01 give about 76–150 and 7–15. This tube-scale estimate places the interfacial resistance well below the radial diffusion resistance for $\alpha_i = 1$ and 0.1, and about an order of magnitude below it for $\alpha_i = 0.01$. It is a scaling estimate, not a substitute for the resolved near-wall gradients of the calculations.

The estimate can be tested directly on a plane channel with the same discretised wall law (fig. 3): 0.1 m long and 4.8 mm high, with both walls condensing, a plug-flow helium carrier, a wall and gas temperature falling linearly from 1000 to 350 K, a constant $D = 10^{-4}\,\mathrm{m^2\,s^{-1}}$ and a 2 s $PbI_2$ release at mid-height. In 13 runs the accommodation coefficient of the $PbI_2$ wall law is varied over six decades, which moves Da, defined by eq. (14) with the half-height $h = 2.4$ mm in place of $R_t$, from $9.63 \times 10^{-4}$ to $1.93 \times 10^3$ at the onset temperature. The normalised wall flux $\hat{J} = Jh/[D(\bar{c} - c_{\mathrm{eq}})]$, with $\bar{c}$ the cross-section mean, equals Da within 0.3% for $\mathrm{Da} \le 0.01$, where the cross-channel profile is flat. For $\mathrm{Da} \gtrsim 300$ it approaches $\pi^2/4 = 2.47$, the diffusion limit of developed plug flow between absorbing plates, within 0.8%, and between the two limits it follows the series form $1/(1/\mathrm{Da} + 4/\pi^2)$ within 5%. Over the same range the deposited fraction of the release rises from 0.3% to 99.5%, the deposit centroid moves upstream from 85.4 to 74.3 mm and the 10–90% width halves from 24 to 12 mm. The 1% onset temperature stays at 561 K within 0.1 K, because the saturation point, not the exchange rate, fixes it. Doubling the wall-normal resolution changes these quantities by at most 0.11%. In the channel the production coefficients give $\mathrm{Da} = 1.6 \times 10^3$–$2.6 \times 10^3$ ($\alpha_i = 1$), 86–135 (0.1) and 8.2–12.9 (0.01) between 400 and 1000 K, close to the tube values above. Relative to $\alpha_i = 1$, the deposit profile changes by 1.4% for $\alpha_i = 0.1$ and by 14% for $\alpha_i = 0.01$, where the wall flux is 20% below the diffusion limit. The production coefficients therefore lie on the diffusion-limited plateau ($\alpha_i = 1$ and 0.1) or at its edge ($\alpha_i = 0.01$). The channel has a plug-flow carrier, a linear temperature ramp and a constant diffusivity; it shows that the discretised wall law reproduces both limits and the transition between them, and where the production coefficients sit on that curve, but it does not vary the interfacial conductance in the column calculations.

The same regime is visible inside the column. Figure 4 shows the $PbI_2$(g) saturation ratio $S = x/x_{\mathrm{eq}}(T_w)$, the mole fraction divided by its equilibrium value at the local wall temperature, in the pure $PbI_2$ steel column at 60 s (production case). The deposit begins where the flow-weighted mean first reaches saturation, at 35.7 cm and 372 °C. Between 35.7 and 41.8 cm (372–266 °C), which holds 99.6% of the deposit, the wall cell stays at $S \le 1.009$, whereas the flow-weighted mean rises from 1.05 to 1.64 and the axis reaches 2.03; at 40 cm the wall-cell mole fraction is 0.64 of the axis value. The upwind 93,312-cell run of the mesh study (section 5) gives the same picture, with at most $S = 1.008$ in the wall cell and a slightly stronger core supersaturation (1.77 flow-weighted, 2.26 on the axis). The interface therefore holds the gas next to the wall at equilibrium, and the deposition rate is set by how fast radial diffusion delivers the supersaturated core vapour. The core supersaturation is also where homogeneous nucleation, which the model omits, could start (section 8.3).

Two omitted effects can be bounded with the same inputs, the digitised wall profiles and the 101,325 Pa outlet pressure. The first is buoyancy. Neither experimental paper states the column orientation. For developed laminar flow with a linearly varying wall temperature, the wall-to-bulk temperature difference is $(11/192)\,Ud^2|\mathrm{d}T_w/\mathrm{d}x|/a$, with $a$ the thermal diffusivity of helium; it stays below 2.6 K. A Grashof number based on the axial temperature change over one diameter, $\mathrm{Gr} = gd^4|\mathrm{d}T_w/\mathrm{d}x|/(T\nu^2)$, stays below 2.2, and below 1.4 where the wall temperature lies between 400 and 750 K. Because Re is also of order one, the Richardson number $\mathrm{Gr}/\mathrm{Re}^2$ reaches 0.15 (pure $PbI_2$) to 0.78 (silica) in that range, and about 1.2 in the steepest cold section of the silica profile. Compositional buoyancy at the largest computed trace mole fraction, $9.2 \times 10^{-5}$ evaluated with the molar mass of $BiI_3$, gives $\mathrm{Gr}/\mathrm{Re}^2 \le 0.32$. Richardson

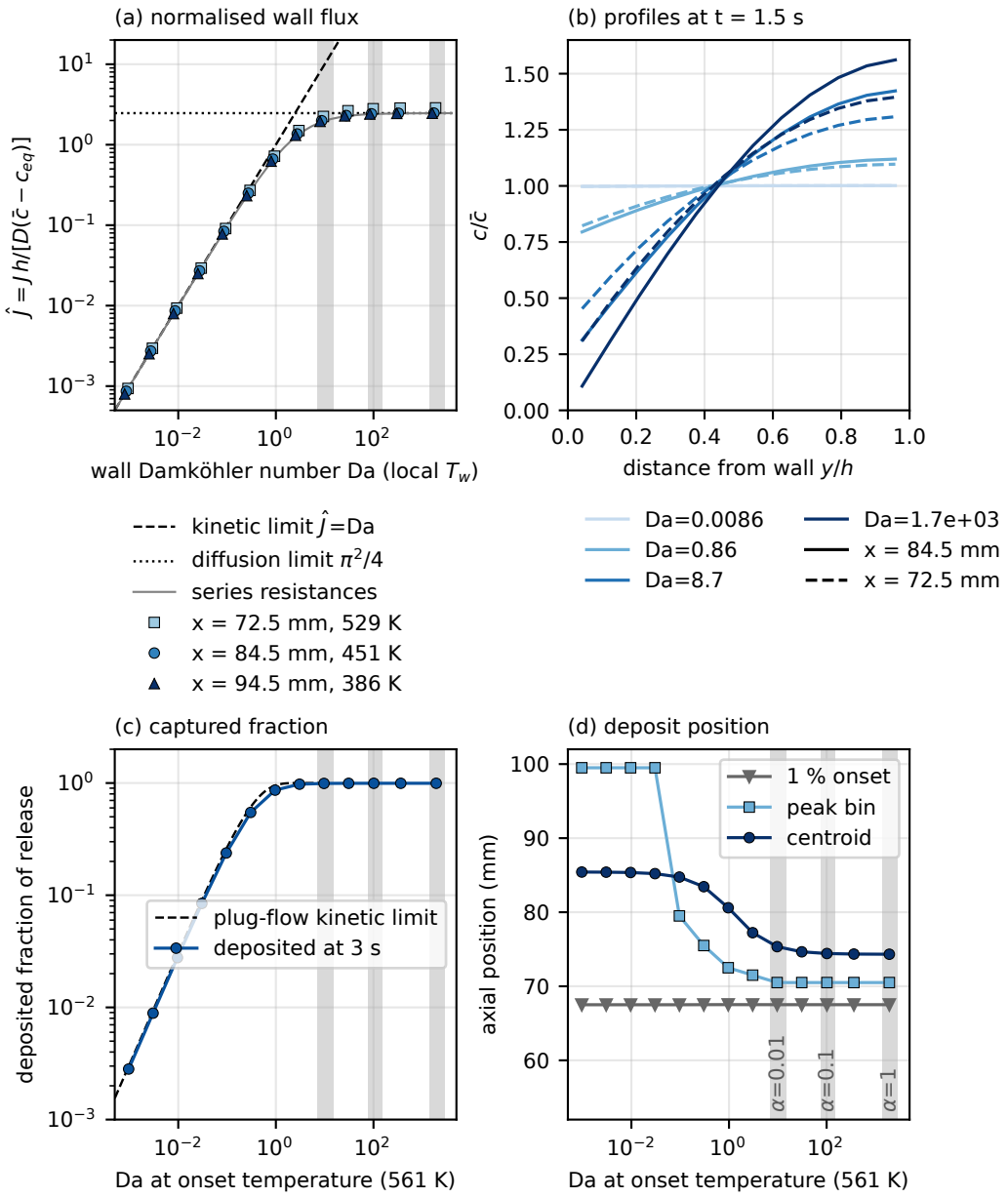



Figure 3: Wall Damköhler sweep on a plane channel with the half-cell wall law ($PbI_2$, plug-flow helium, wall temperature 1000–350 K, $D = 10^{-4}\,\mathrm{m^2\,s^{-1}}$). (a) Normalised wall flux during the release ($t = 1.5$ s) against the local Da at three axial stations, with the kinetic limit $\hat{J} = \mathrm{Da}$, the plug-flow diffusion limit $\pi^2/4$ and the series-resistance curve. (b) Wall-normal concentration profiles at 72.5 mm (dashed) and 84.5 mm (solid) for four values of Da. (c) Fraction of the release deposited at 3 s, with a plug-flow kinetic estimate that has no fitted parameter. (d) Onset, peak-bin and centroid positions of the deposit; for $\mathrm{Da} < 0.05$ the peak bin is the outlet bin, because the deposit reaches the end of the channel. In (c) and (d) Da is evaluated at the onset temperature, 561 K. Grey bands: the tube Damköhler numbers of the production cases for $\alpha_i = 1$, 0.1 and 0.01 (section 4.2).

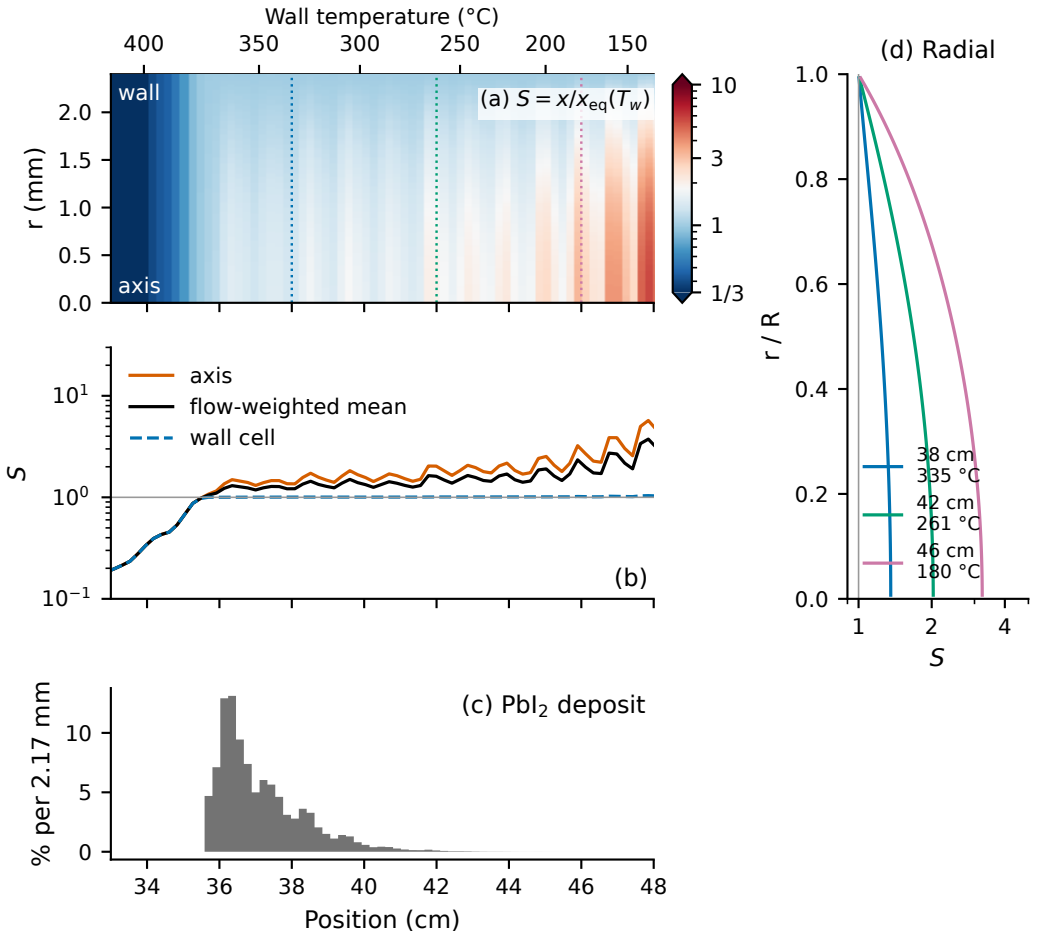



Figure 4: $PbI_2$(g) saturation ratio in the pure $PbI_2$ steel column at 60 s (production case: limited-linear advection, 41,472 cells). (a) $S = x/x_{eq}(T_w)$ with $x$ the $PbI_2$ mole fraction and $x_{eq}$ the equilibrium mole fraction at the local wall temperature (top axis); the radius is stretched. (b) $S$ on the axis, flow-weighted over the section and in the wall cell. (c) Wall deposit per axial cell. (d) Radial profiles at the dotted stations of (a). The ripples along $x$ follow the slope changes of the digitised wall temperature, to which the exponential equilibrium pressure is sensitive.

numbers of order 0.1–1 do not show that buoyancy is negligible: in a horizontal column, buoyancy-driven secondary flow may be significant locally, in particular in the steep cold sections where deposition occurs. The frozen carrier does not represent it, and with the orientation unknown its effect on the deposition profiles is an unquantified limitation of the model. The second effect is thermal (Soret) diffusion. Taking a thermal diffusion factor $\alpha_T = 1$ as a generous bound, the Soret drift velocity $D_i \alpha_T |\mathrm{d}T_w/\mathrm{d}x|/T$ is at most 2.2% of the carrier velocity. The radial Soret flux, relative to a Fickian flux $x_i D_i / R_t$ towards the wall, is at most $\alpha_T \Delta T_r / T \leq 0.7\%$, with $\Delta T_r$ the wall-to-bulk difference above. Both are small compared with the factor-of-two diffusivity variations examined in section 7.

### 4.3 Discretisation and coupling

The transient species equations use implicit Euler time integration and corrected Gauss linear diffusion. The production cases (the 15 baselines and four silica temperature-coordinate cases) use the bounded OpenFOAM scheme `Gauss limitedLinear 1` [18, 19] for advection on 41,472 cells. It blends linear and upwind interpolation through a TVD-type flux limiter, which reverts to upwinding near extrema and steep gradients. The 64 follow-up cases and the time-step comparisons keep the first-order upwind advection of the original campaign on 18,432 cells, and the follow-up sensitivities are always reported as differences from their own upwind baselines on that mesh. The diffusivity and accommodation variations were also repeated with the production scheme on 41,472 cells for two baselines, pure $PbI_2$ and steel Q2 at 1% metal saturation (8 runs, table 8). The scheme and mesh study adds the van Leer limiter [20] (`Gauss vanLeer`) as a second bounded scheme. All nine transported gases use the same advection scheme

within a case. The baseline time step is 2 ms. Wall exchange is assembled implicitly with nonnegative source and sink coefficients and shared inventory bounds. Configured species correctors and active-bound guards reconcile the final gas solution with the realised wall transfer. Gas re-speciation then redistributes species under the local element constraints. The sequence is first order in time; a conservative equilibrium projection alone does not remove splitting or transport error. The baseline configuration requests three species correctors with a $10^{-16}$ tolerance and up to eight bound guards with a $10^{-13}$ guard tolerance; the effect of this stopping criterion is tested in section 5.2.

Every gas has the baseline diffusivity $D_i = 10^{-4}\,\mathrm{m^2\,s^{-1}}$. Three transport alternatives halve or double this value for all gases, or replace only the $PbI_2$ coefficient with a temperature- and pressure-dependent $PbI_2$–He estimate whose collision parameters are themselves estimates (supplementary material, Sec. S5). Accommodation variations set $\alpha = 0.1$ or 0.01 for the two reversible iodide pairs and keep the baseline metal-channel coefficients.

### 4.4 Completed case matrix

The 15 baseline cases are Q1 for each available configuration, plus Q2 for the two LBE columns at $\beta \in \{0.001, 0.01, 0.1\}$ with two thermodynamic branches. The second branch adds a constant 29 kJ/mol to the standard Gibbs energy of BiI(g); it is a sensitivity surrogate, not a second dataset. The 64 follow-up cases vary mesh and time step, diffusivity, accommodation, source position, Q2 metal loading, flow reference, silica temperature coordinate and a fixed-temperature flow stop; they are a structured set of conditional comparisons, not a factorial uncertainty study. Six further sets complete the evidence: the 19 production re-runs (section 5), the 8 production-scheme sensitivities, a scheme and mesh study of two cases up to 93,312 cells with three further upwind runs on that mesh, four full-run corrector checks, ten runs of five single-species experiments of Lobresco et al. [6] on two meshes (section 6.3), and the 13 channel runs of the Damköhler sweep with 5 refined repeats (section 4.2). All these calculations use the same solver executable, identified by its SHA-256 in the saved reports. Table S1 and Sec. S1 of the supplementary material list every case, its variation and its reference case.

## 5 Numerical verification and resolution limits

This section verifies the wall-exchange discretisation on a problem with a reference solution, then reports material conservation and the time-step, iteration, mesh and scheme dependence of the iodine calculations.

### 5.1 Verification of the wall-exchange discretisation

The half-cell series law of eq. (5) is verified on an extended Graetz problem with a Robin wall condition [2]: fully developed laminar flow in a pipe, axial diffusion included, and a constant first-order wall uptake $-\rho D\,\partial Y/\partial r = \rho k_w Y$ at $r = R_t$. Far from the inlet and the outlet every cross-sectional measure decays as $\exp(-\kappa x)$, and the decay rate depends only on the Péclet number $\mathrm{Pe}_R = U_b R_t/D$ and the Biot number $\mathrm{Bi} = k_w R_t/D$, the analogue of the wall Damköhler number of eq. (14). The reference $\kappa R_t$ for $\mathrm{Pe}_R = 5$ and $\mathrm{Bi} = 1$ and 50 is the smallest positive eigenvalue of the radial problem, computed by a power series in extended precision, by shooting and by Chebyshev collocation; the three agree within a relative $1.69 \times 10^{-12}$.

The test uses the solver's own wall-exchange implementation with a constant uptake velocity $k_w$ ($R_t = 2.4$ mm, $T = 500$ K, $p = 1$ atm, $D = 7.6963 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$, a pipe of length $24R_t$). The meshes are 2° wedges with exactly axisymmetric cell metrics and $N_r = 32$, 64, 128 and 256 uniform radial cells, $\Delta x = 2\Delta r$, and the parabolic carrier enters through exact face fluxes. Steady states are obtained by implicit Euler steps with a very large time step, and $\kappa$ is fitted to the computed gas line density in a window chosen from the spectrum. Two wall laws are compared: the half-cell series law and the kinetic law applied directly to the cell-centre value (no half-cell resistance), which is the form of a volumetric first-cell sink. Every fitted $\kappa$ equals the decay rate of the developed discrete mode of the documented scheme within $7.09 \times 10^{-9}$, and moving the fit window or changing the outlet condition changes it by less than $10^{-7}$, so the differences below are discretisation errors. All 27 acceptance criteria of the study pass (records, provenance and regeneration: Sec. S7 of the supplementary material).

Table 2 gives the relative error of $\kappa$ on each mesh, the observed order and the grid convergence index $\mathrm{GCI}_{\mathrm{fine}}$ of Celik et al. [21] on $N_r = 64/128/256$. The half-cell law converges at order 1.99–2.00 with $\mathrm{GCI}_{\mathrm{fine}} \le 1.89 \times 10^{-4}\%$, and its Richardson-extrapolated decay rate matches the reference within $9.47 \times 10^{-9}$. The cell-centre law converges at first order (1.004–1.008) with $\mathrm{GCI}_{\mathrm{fine}} = 0.155$–$0.341\%$. With bounded SuperBee advection the observed orders are 2.00–2.01 and 1.01–1.02, respectively (supplementary material, Table S4). On the coarsest mesh the cell-centre law errs by 1–2% in the decay rate, the half-cell law by at most $10^{-4}$.

Table 2: Graetz–Robin verification ($\mathrm{Pe}_R = 5$): relative error $\kappa/\kappa_{\mathrm{ref}} - 1$ of the fitted decay rate on $N_r$ radial cells, observed order $p$ and $\mathrm{GCI}_{\mathrm{fine}}$ [21] on $N_r = 64/128/256$, Gauss linear advection, zero-gradient outlet. Half-cell series: eq. (5); cell-centre value: kinetic law without the half-cell resistance.

| Bi | Wall law | $N_r = 32$ | 64 | 128 | 256 | $p$ | GCI (%) |
|---|---|---|---|---|---|---|---|
| 1 | Half-cell series | $9.7 \times 10^{-5}$ | $2.4 \times 10^{-5}$ | $6.1 \times 10^{-6}$ | $1.5 \times 10^{-6}$ | 2.001 | 0.000189 |
| 1 | Cell-centre value | $1.0 \times 10^{-2}$ | $5.0 \times 10^{-3}$ | $2.5 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | 1.008 | 0.155 |
| 50 | Half-cell series | $-5.0 \times 10^{-5}$ | $-1.3 \times 10^{-5}$ | $-3.2 \times 10^{-6}$ | $-7.9 \times 10^{-7}$ | 1.992 | 0.0001 |
| 50 | Cell-centre value | $2.2 \times 10^{-2}$ | $1.1 \times 10^{-2}$ | $5.5 \times 10^{-3}$ | $2.7 \times 10^{-3}$ | 1.004 | 0.341 |

This test verifies the discretisation of the wall exchange and its coupling to advection and diffusion for a linear wall law at $\mathrm{Bi} \leq 50$. It does not verify the nonlinear HKS law with tabulated equilibrium pressures, the gas equilibrium or the wall inventories, and it does not show that the full iodine calculations are mesh-converged; their resolution is assessed separately below. The iodide Damköhler numbers of section 4.2 also exceed the tested Biot numbers, and the meshes of the iodine calculations have cell aspect ratios far from the $\Delta x = 2\Delta r$ of the test.

### 5.2 Material balance and solver diagnostics

All 15 runnable baseline cases, all 64 follow-up cases, the 19 production cases and the four full-run corrector cases completed and passed their configured numerical acceptance checks; the scheme and mesh runs completed with the same ledgers. This PASS classification is not an experimental-validation classification. For element $k$ the raw material residue is

$$R_k = N_k^{\mathrm{gas}} + N_k^{\mathrm{wall}} + N_k^{\mathrm{sample}} + N_k^{\mathrm{net\,out}} + N_k^{\mathrm{removed}} - N_k^{\mathrm{initial}} - N_k^{\mathrm{released}}, \tag{15}$$

where $N_k$ are molar atom inventories and outward transport is signed; the linear-solve defect and clamping corrections are recorded separately and are not inserted into $R_k$. Divided by the supplied material and maximised over steps and elements, the largest raw residue is $1.41 \times 10^{-12}$ for the 19 production cases and $8.85 \times 10^{-13}$ for the 60 source-driven follow-up cases. The calculations conserve the represented elements to round-off; this says nothing about physical accuracy or discretisation error. The ledger terms, the equilibrium-solver diagnostics, the startup sign flags and the trace-loading bound are given in Sec. S2 of the supplementary material.

Some paired wall-coupling iterations do not meet the requested $10^{-16}$ outer criterion. Four cases were therefore repeated over the full 60 s with eight instead of three correctors (18,432 cells, upwind). The iodine profiles of pure $PbI_2$ and of the steel Q2 case at 0.1% saturation change by $1.09 \times 10^{-13}\%$ and $4.89 \times 10^{-13}\%$, that is, at round-off; the steel case at 1% saturation changes by 0.021%, and the same case with the shifted BiI data by $1.10 \times 10^{-7}\%$. The requested tolerance lies below the attainable residual floor (Sec. S2), and the iteration error of the reported profiles is negligible next to the time-step and spatial errors below.

### 5.3 Time-step, mesh and scheme dependence

For profiles on the common 1 cm bins, let $a_b$ be the deposited iodine atoms in bin $b$ for a comparison case and $a_b^0$ the corresponding reference inventory. We report the profile change $E_{L1}$ and the total change $\Delta N_I$:

$$E_{L1} = 100 \frac{\sum_b |a_b - a_b^0|}{\sum_b a_b^0}, \qquad \Delta N_I = 100 \frac{\sum_b a_b - \sum_b a_b^0}{\sum_b a_b^0}. \tag{16}$$

Both measures retain changes in amount. Source, physical coefficients, equilibrium engine and executable are fixed within each numerical comparison.

Table 3 separates time-step from mesh and scheme effects; for successive mesh levels the reference $a_b^0$ is the finer level. Reducing the time step from 2 to 1 ms on 18,432 cells changes the iodine profile by 0.0043–0.3228%. With the upwind scheme of the original campaign, the 18,432 to 41,472-cell differences of the four follow-up cases are 2.06–10.58%, while total deposited iodine changes by less than 0.014% and the centroid moves by at most 0.0652 cm. Integral quantities are thus stable while local profile shapes still differ appreciably.

The scheme and mesh study (table 3 and fig. 5) addresses the shifted-BiI steel case, which had the largest upwind change between 18,432 and 41,472 cells. With limited-linear advection its iodine profile changes by 15.35, 8.95 and 1.98% over the three refinements to 93,312 cells; with upwind, by 10.47, 10.58 and 2.13%. Pure $PbI_2$ changes by 5.70 and 2.15% with limited-linear, against 6.57 and 5.06% with upwind. The van Leer scheme differs from limited-linear

Table 3: Iodine profile changes $E_{L1}$ (percent) between successive mesh levels at 2 ms, relative to the finer level, for three advection schemes, and the effect of halving the time step on 18,432 cells with upwind advection. All steel cases are Q2; the shifted-BiI case uses 1% metal saturation. The 41,472→93,312 upwind values of pure $PbI_2$ and the two unshifted steel cases come from separate runs of the follow-up campaign. A dash marks a combination that was not run.

| Case | Scheme | 8,192→18,432 | 18,432→41,472 | 41,472→93,312 | 2→1 ms |
|---|---|---|---|---|---|
| Pure $PbI_2$ | upwind | 6.57 | 5.06 | 3.68 | 0.0043 |
| | limitedLinear | 5.70 | 2.15 | – | – |
| | van Leer | 4.94 | 2.23 | – | – |
| Steel, shifted BiI | upwind | 10.47 | 10.58 | 2.13 | 0.1536 |
| | limitedLinear | 15.35 | 8.95 | 1.98 | – |
| | van Leer | 16.83 | 9.17 | – | – |
| Steel Q2, 0.1% | upwind | 9.02 | 5.26 | 7.30 | 0.2326 |
| Steel Q2, 1% | upwind | 9.72 | 2.06 | 6.68 | 0.3228 |

by at most 2.8% (steel) and 1.8% (pure $PbI_2$) at equal mesh. Between 41,472 and 93,312 cells, the 3 cm smoothed profile of the steel case changes by 0.67% and its total deposit by 0.015%; over all four levels its centroid stays within 0.6 mm and the wall temperature at the $PbI_2$ centroid within 0.9 K.

The successive limited-linear profile changes of the steel case give apparent orders of 1.3 and 3.7, and its iodine total, centroid and width oscillate on both triplets. These orders are inconsistent, so no formal grid convergence index is claimed [21–23]. We quote the measured 41,472 to 93,312-cell change of 1.98% as the spatial uncertainty indicator of this production profile. Extrapolating the last change at the last apparent order would place the 41,472-cell profile about 2.5% from the mesh limit, and assuming second order about 3.6%. These are indicative estimates for one case, not confirmed bounds, and they do not establish mesh independence of local profile shapes in other cases. Upwind runs of all four check cases on 93,312 cells show this directly: from 41,472 to 93,312 cells the iodine profiles of pure $PbI_2$ and of the two unshifted steel cases change by 3.68–7.30%, more than the 2.13% of the shifted-BiI case, while their totals change by less than 0.02%. The shifted-BiI case is therefore not the most mesh-sensitive case at the finest step, and the 1.98% bounded-scheme change does not bound the spatial error of the other production profiles; the bounded scheme was refined to 93,312 cells for this case only. The three-grid scalar checks of the earlier upwind triplets are in Sec. S3 of the supplementary material.

At equal mesh, upwind differs from limited-linear by 20.8, 14.1, 11.3 and 7.7% for the steel case on the four levels and by 9.0–13.2% for pure $PbI_2$, much more than the two bounded schemes differ from each other. This is consistent with the local estimate $D_{\mathrm{num}} \simeq |u_x|\Delta x/2$, which is comparable to or greater than the prescribed $10^{-4}\,\mathrm{m^2\,s^{-1}}$ in parts of the carriers. Upwind numerical diffusion is therefore the dominant spatial error of the original campaign, and it is why halving the time step could not establish convergence of profile widths. Across the 19 production cases, the production profiles differ from the 18,432-cell upwind profiles by 9.1–22.3%. The follow-up sensitivities of section 7 remain upwind differences on 18,432 cells, taken against baselines with the same numerical error.

### 5.4 Binned maxima and observation width

A deposition maximum picked on a 1 cm grid is fragile when adjacent bins have similar heights. In pure $PbI_2$ the 18,432-cell upwind maximum is at 37 cm and 350.7 °C, while the 36 cm bin is only about 0.05% lower; on 41,472 cells it moves to 36 cm and 366.5 °C although the centroid shifts by only $-0.0585$ cm. In the limited-linear steel sequence the $PbI_2$ maximum alternates between two nearly equal bins, 50 and 51 cm (306.3, 288.1, 306.7 and 288.1 °C on the four levels), while the 3 cm smoothed maximum stays at 51 cm and 288.0–288.1 °C and the centroid temperature at 290.6–291.5 °C. A one-bin switch can change the reported peak temperature by tens of kelvin without a comparable displacement of the profile, so table 4 also reports the smoothed and centroid values. Conversely, comparing profiles on wider bins does not remove the mesh sensitivity: for the shifted-BiI case the original upwind fine/medium difference on 4 cm bins ranges from 0.52% to 10.02% depending on the bin origin (Sec. S4 of the supplementary material, which also gives a transport distance between the binned profiles).

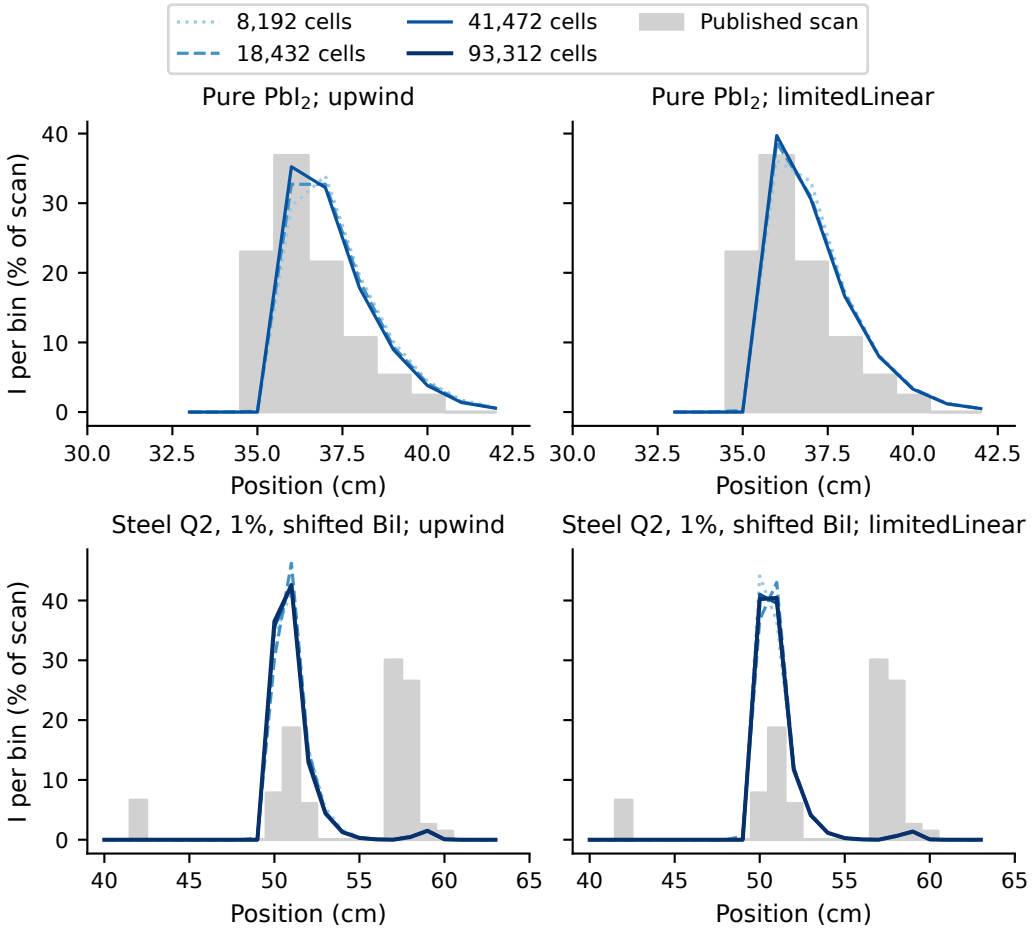



Figure 5: Mesh sequences of the iodine deposition profile for pure $PbI_2$ (top) and the shifted-BiI steel case at 1% metal saturation (bottom), with first-order upwind (left) and limited-linear (right) advection. Each curve is normalised over the common scanned interval so that shapes can be compared; the unnormalised differences are given in table 3. Grey histograms are the reconstructed experimental iodine scans [1] and include any KI contribution.

# 6 Conditional comparison with deposition measurements

This section compares the calculations, mainly the original Q1 cases, with the reconstructed deposition scans of Liu et al., and the single-species model with the experiments of Lobresco et al. The comparisons are conditional on the reconstructed inputs and assumptions of sections 2 and 3.

## 6.1 Comparison measures

Against experiment we use reconstructed relative iodine yields per 1 cm bin. Let $e_b$ be the experimental percentage and let

$$m_b = (100 - y_{\mathrm{KI}}) \frac{a_b}{\sum_{\text{full column}} a_b} \tag{17}$$

be the modelled percentage, where the stated excluded KI yield is $y_{\mathrm{KI}} = 0$, 6.6 or 7.0. Over the common scanned bins $\mathcal{B}$, we report the raw discrepancy

$$L_{\mathrm{raw}} = \sum_{b \in \mathcal{B}} |m_b - e_b| \quad \text{(percentage points)}, \tag{18}$$

Table 4: Q1 iodide maxima on the production mesh (41,472 cells, limited-linear advection), compared with the published peaks. Subscript bin: 1 cm binned maximum; subscript 3: maximum of the 3 cm smoothed profile; $T_c$: wall temperature at the deposit centroid. Positions are in cm and temperatures in °C. Experimental temperature ranges are those printed by Liu et al. [1]; they are not statistical confidence intervals for model error.

| Case | Species | $x_{exp}$ | $x_{bin}$ | $x_3$ | $T_{exp}$ | $T_{bin}$ | $T_3$ | $T_c$ |
|---|---|---|---|---|---|---|---|---|
| Pure $PbI_2$, steel | $PbI_2$ | 36 | 36 | 37 | 368±18 | 366.5 | 350.5 | 348.7 |
| LBE, steel | $PbI_2$ | 51 | 51 | 52 | 288±18 | 288.1 | 269.2 | 274.5 |
| LBE, steel | $BiI_3$ | 57 | 57 | 58 | 154±14 | 166.9 | 138.8 | 158.1 |
| LBE, silica | $PbI_2$ | 51 | 56 | 56 | 306±24 | 287.8 | 287.8 | 288.8 |
| LBE, silica | $BiI_3$ | 59 | 63 | 64 | 127±39 | 152.8 | 119.8 | 147.1 |

which keeps the experimental digitisation sum and the model's explicit KI exclusion. A secondary shape diagnostic normalises both profiles over $\mathcal{B}$, $\widehat{m}_b = m_b / \sum_{\mathcal{B}} m_b$ and $\widehat{e}_b = e_b / \sum_{\mathcal{B}} e_b$, and computes the overlap

$$O = 100 \sum_{b \in \mathcal{B}} \min(\widehat{m}_b, \widehat{e}_b) = 100 \Big[ 1 - \tfrac{1}{2} \sum_{b \in \mathcal{B}} |\widehat{m}_b - \widehat{e}_b| \Big]. \tag{19}$$

Identical binned shapes give 100% and disjoint shapes zero. $O$ discards differences in capture and total yield, and its experimental normalisation still contains KI. $L_{raw}$, $O$ and the simulation-to-simulation change $E_{L1}$ (eq. (16)) have different denominators and are not interchangeable percentages of model accuracy.

### 6.2 Q1 peaks and complete profiles

Table 4 compares the five available Q1 iodide maxima with the printed peak temperatures. Besides the 1 cm binned maximum it lists the maximum of the 3 cm smoothed profile ($x_3$, $T_3$) and the wall temperature at the deposit centroid ($T_c$). All five binned peak temperatures lie within the printed ranges, but the other two measures do not always. For pure $PbI_2$ the binned maximum (36 cm, 366.5 °C) matches the published 36 cm and 368 °C; the smoothed maximum (350.5 °C) lies at the lower limit of the printed $368 \pm 18$ °C, and the centroid temperature (348.7 °C) lies 1.3 K below that 350 °C limit. For steel $PbI_2$ the binned value is 288.1 °C, the centroid temperature (274.5 °C) lies near the lower limit of $288 \pm 18$ °C, and the smoothed maximum (269.2 °C) lies 0.8 K below its 270 °C limit. The steel $BiI_3$ smoothed maximum (138.8 °C) falls just below its range, while the centroid temperature (158.1 °C) lies inside it. Whether a computed peak temperature agrees with a printed range therefore depends on which measure of the peak is used (section 5). Steel LBE places both iodide maxima at the measured positions, whereas silica LBE places the $PbI_2$ and $BiI_3$ maxima 5 and 4 cm downstream; agreement with a printed temperature does not establish the spatial location.

Figure 6 shows the complete profiles. The raw discrepancies $L_{raw}$ are 46.33, 52.97 and 183.84 percentage points for pure $PbI_2$, steel LBE and silica LBE (53.90, 53.45 and 181.84 on the 18,432-cell upwind mesh), and the corresponding normalised overlaps are 76.97%, 72.13% and 6.13% (72.93%, 71.88% and 7.20%). The silica case has a large profile discrepancy even though both printed temperature ranges contain its calculated binned maxima.

The Q1 steel $BiI_3$ share of deposited iodide iodine is 65.34%, close to the measured iodide-only share of $61.3/(61.3 + 32.1) = 65.63\%$; for silica the values are 28.68% and $27/(27 + 66) = 29.03\%$. Because the Q1 source masses were inferred from the same deposits, these agreements cannot serve as independent tests of gas speciation or LBE release. Similarly, the source-driven calculations capture 91.28–98.20% of the supplied iodine on the wall by 60 s, which says nothing about the experimental three-hour capture.

### 6.3 Single-species comparison: Lobresco et al.

The five $PbI_2$ thermosublimatography experiments of Lobresco et al. [6] vary the helium flow (5, 106 and 540 mL/min) and the duration (6, 60 and 300 s) with 1.37–2.23 mg samples in a 4.8 mm steel column. Unlike the Liu et al. cases, they do not require a deposit-derived source. That study also reports the deposition temperatures of two of its own CFD models: a temperature-based model calibrated against one reference experiment, and a vapour-pressure-based Hertz–Knudsen–Schrage model with a near-wall interaction layer and an effective rate multiplier $C_e$. We ran the five cases without adjusting any parameter to these data.

**Comparison region.** The deposition temperature is defined as in that study: the wall temperature at the first position where the deposit exceeds 1% of its peak, linearly interpolated between 1 cm points. We apply this rule to the measured profiles from their first digitised point, and to our computed deposits downstream of the end of the sample, in all five

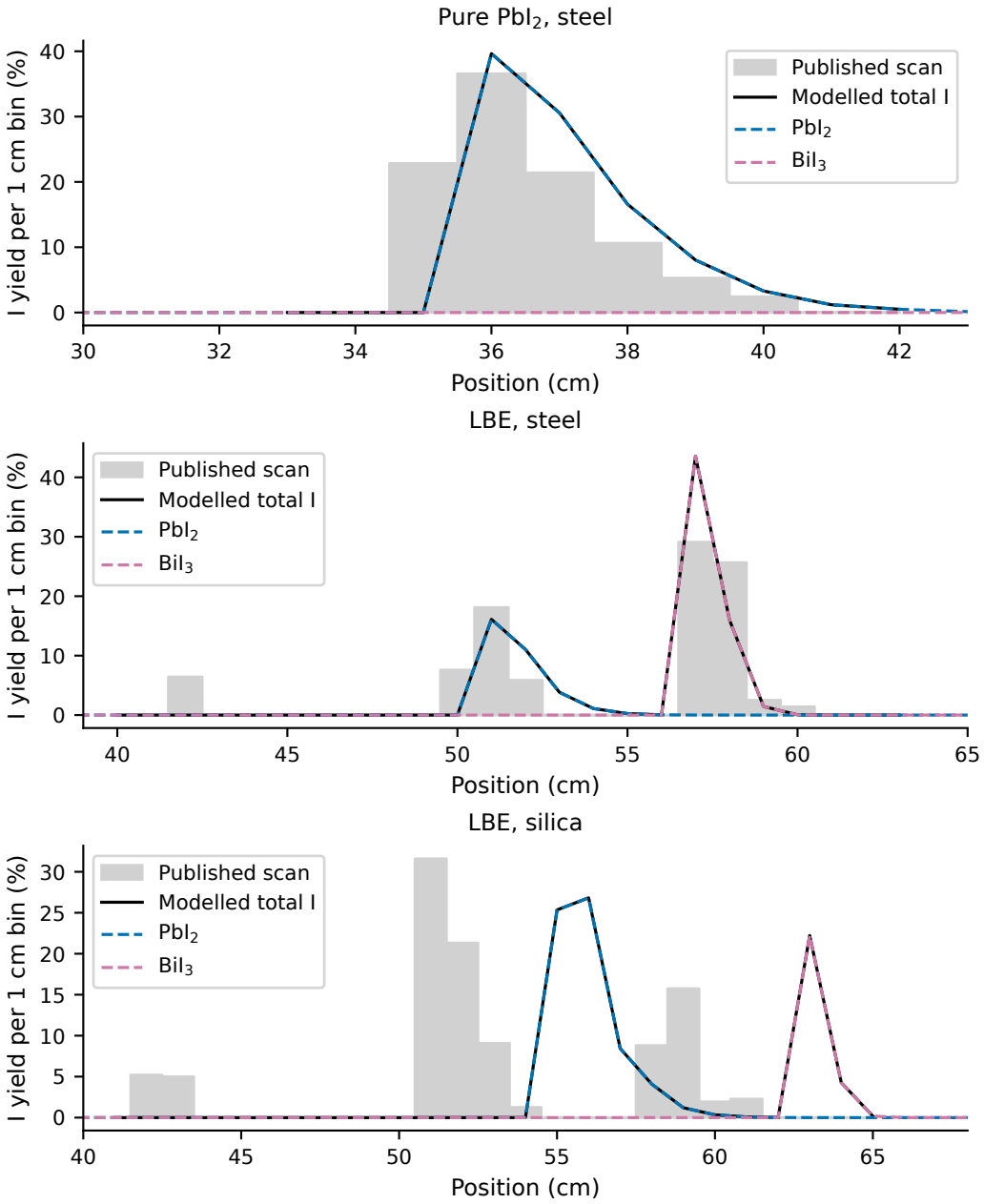



Figure 6: Production Q1 iodine profiles (41,472 cells, limited-linear advection) compared with the digitised published scans (grey histograms) [1]. Solid black lines are total modelled wall iodine; dashed lines show its two iodide contributions. Model totals are normalised to 100%, 93.4% and 93.0% for pure $PbI_2$, steel LBE and silica LBE, respectively, to expose the omitted KI yield; experimental bars keep their digitised values.

cases. We take the rule to locate the onset of the main deposit formed by transport down the gradient, and the measured profiles contain only that deposit: the measured T5_60 profile, digitised from 2 cm, has no deposit above the threshold upstream of 37 cm. For four cases the restriction makes no difference. At 5 mL/min, however, computed vapour diffuses upstream against the slow carrier through the zero-total-flux inlet condition (below), and 1.2% of the T5_60 deposit forms upstream of the sample, near the cold inlet. Applied from the inlet, the rule picks this deposit at 548 K instead of the main-deposit onset at 836 K, and the RMS deviation of our five onsets becomes 135 K instead of 18 K. The T5_60 value, and with it the RMS, therefore depends on the choice of comparison region; the computed upstream deposit is itself a discrepancy with the measurement.

Table 5: Deposition-onset temperatures for the five $PbI_2$ experiments of Lobresco et al. [6]: flow $\dot{V}$ (mL/min, standard), duration $t$ (s), measured onset $T_{\mathrm{exp}}$ (K, their Table 2), and the onset of this work and of their temperature-based and vapour-pressure-based models minus $T_{\mathrm{exp}}$ (K; the latter two from their Table 2). $\Delta x_{\mathrm{pk}}$ is our deposit peak minus the measured peak (cm), and $L_1/n_0$ the summed absolute difference of our deposit and the digitised measured profile divided by the sample amount. This work: 22,896 cells, deposit at the end of the experiment, onset rule applied downstream of the sample in every case.

| Case | $\dot{V}$ | $t$ | $T_{\mathrm{exp}}$ | This work | Temp.-based | Vap.-press. | $\Delta x_{\mathrm{pk}}$ | $L_1/n_0$ |
|---|---|---|---|---|---|---|---|---|
| T106_6 | 106 | 6 | 814 | +26 | −39 | +27 | −4 | 0.89 |
| T106_60 | 106 | 60 | 723 | +19 | −10 | +37 | −2 | 0.66 |
| T106_300 | 106 | 300 | 667 | +16 | −1 | +34 | −2 | 0.88 |
| T5_60 | 5 | 60 | 847 | −11 | −60 | −50 | −1 | 0.37 |
| T540_60 | 540 | 60 | 700 | −14 | −30 | −15 | −1 | 0.47 |
| RMS | | | | 18 | 35 | 35 | | |

**Inputs.** Flows (standard volumetric rates at 273.15 K and 101,325 Pa), durations and sample amounts are those of Table 1 of that study. The wall temperature of each flow rate, the measured profiles, the sample position and both published model curves were digitised from its Figs. 13–14 (CC BY 4.0); the 1% rule applied to the digitised measured profiles reproduces the published experimental deposition temperatures within 1.0 K. The model is that of sections 3 and 4 reduced to one species: an expanding laminar helium carrier computed with the wall temperature, HKS exchange with $C_e = \alpha = 1$ in series with the half-cell diffusion resistance, the $PbI_2$–He diffusivity correlation of that study, our own phase-resolved $PbI_2$ vapour-pressure table, an inlet condition of zero total species flux, and an evaporating sample of the Table 1 amount at the digitised position, removed at the end of the experiment. Mesh spacings follow the production and 18,432-cell meshes (22,896 and 10,176 cells), with limited-linear advection; details, including a wall-temperature table of the T-Flows test case that differs from the published curves, are in Sec. S6 of the supplementary material. Our calculation thus differs from both published models in more than the wall treatment: they prescribe a fully developed laminar velocity profile, take the $PbI_2$ vapour pressure from a gas–solid equilibrium calculation, and represent the source as a timed inlet release (temperature-based model) or a volumetric solid inventory (vapour-pressure model).

**Results.** Table 5 and fig. 7 summarise the comparison. Our onset temperatures differ from the measurements by +26, +19, +16, −11 and −14 K, an RMS deviation of 18 K, against 35 K for the calibrated temperature-based model and 35 K for the vapour-pressure-based model. The two meshes agree within 0.2 K. The measured decrease of the onset temperature with duration at 106 mL/min is reproduced with a warm offset of 16–26 K; the onsets at 5 and 540 mL/min are 11–14 K too cold. Three of the five deviations exceed the stated ±15 K experimental temperature uncertainty. The profiles agree less well than the onsets: the computed peaks lie 1–4 cm upstream of the measured peaks, and the summed absolute profile difference is 0.37–0.89 of the sample amount (table 5).

**Limits of the comparison.** Three limits restrict what the onset agreement can show. First, the comparison-region choice above decides the T5_60 result. Second, the near-source state lies outside the model: next to the sample the computed $PbI_2$ mole fraction reaches 0.33–0.36, which violates the dilute, frozen-carrier assumption (the vapour would alter density, velocity and diffusion), and the boat is not represented. Third, the release history is not constrained. In the model the sample is fully evaporated before the end of even the shortest, 6 s experiment: by then 98.1% of the T106_6 sample is deposited and the rest is gaseous. Lobresco et al. report that complete evaporation was confirmed in the experiments and that the evidence indicates near-complete evaporation within the shortest experiment, which supports release within 6 s. No measurement resolves the release on shorter time scales; the heat-up of the inserted boat and the actual sample surface are unknown, and the computed release rate comes from the non-dilute region where the model does not apply. The experiments also end by stopping the flow, withdrawing the sample and pulling the column out of the furnace, which is not simulated; counting the vapour still in the tube at the end time changes only T5_60 noticeably, to +3 K, because 4.0% of its inventory is still gaseous (RMS 17 K).

The onset comparison is therefore promising but conditional. Carrier, vapour-pressure and source assumptions differ from those of the published models, so the smaller RMS deviation cannot yet be attributed to the wall treatment, and the profile discrepancies above remain. It is an untuned comparison with published measurements and published model results, not a validation. In the channel sweep of section 4.2 the 1% onset does not move when the accommodation coefficient changes by six decades, so this metric reflects mainly where the gas saturates, which depends on the vapour pressure, carrier and source, rather than the interfacial kinetics.

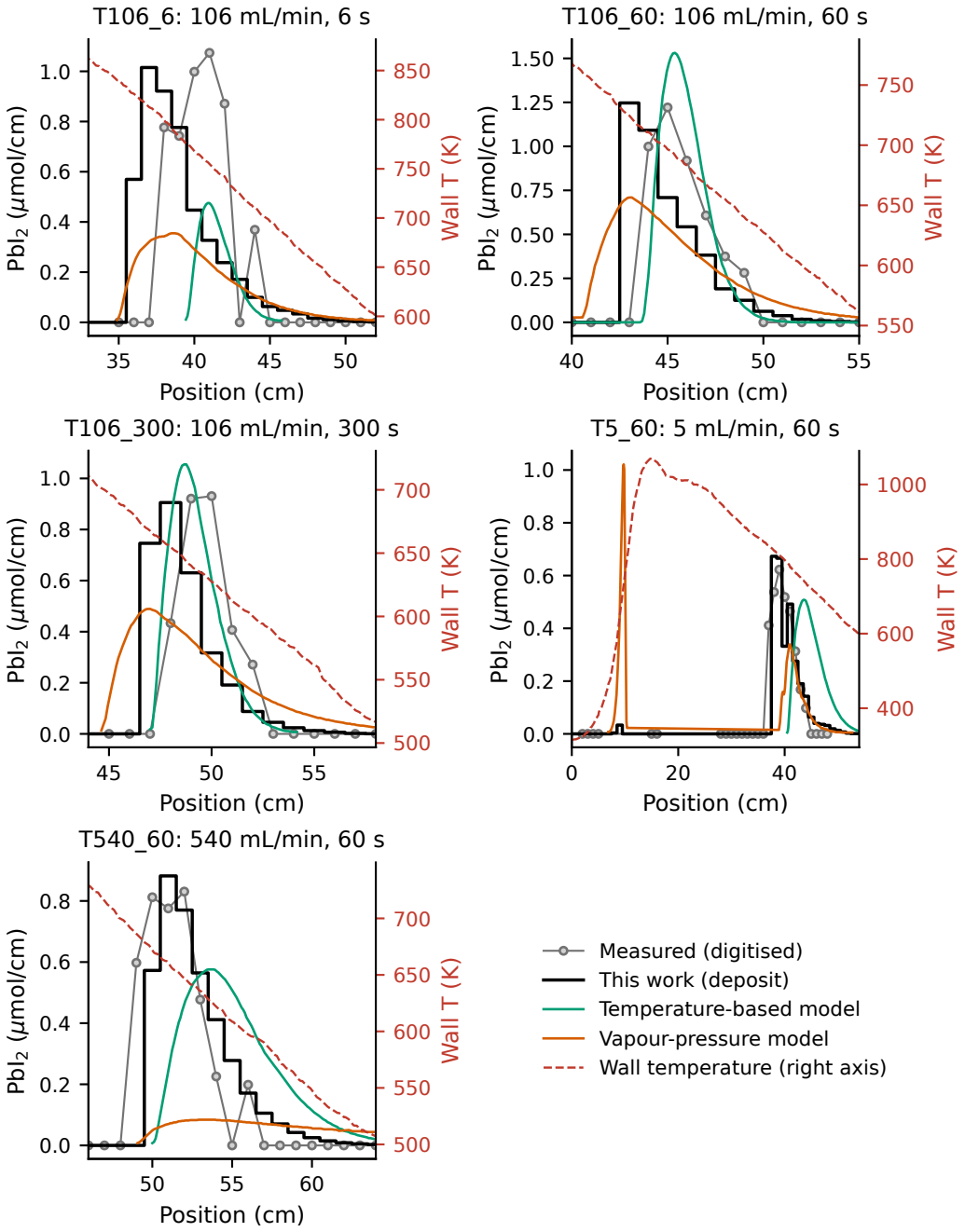



Figure 7: Deposited $PbI_2$ for the five experiments of Lobresco et al. [6]: measured profiles, their temperature-based and vapour-pressure-based model results (all digitised from their Figs. 13–14 and adapted under CC BY 4.0), and the deposit of this work at the end of each experiment (1 cm bins, 22,896 cells). Dashed red lines: digitised wall temperature used as input (right axes).

## 7 Input sensitivity and interpretation

This section examines how source loading, BiI thermodynamic data, the silica temperature coordinate, and transport, kinetic and flow inputs change the computed deposits.

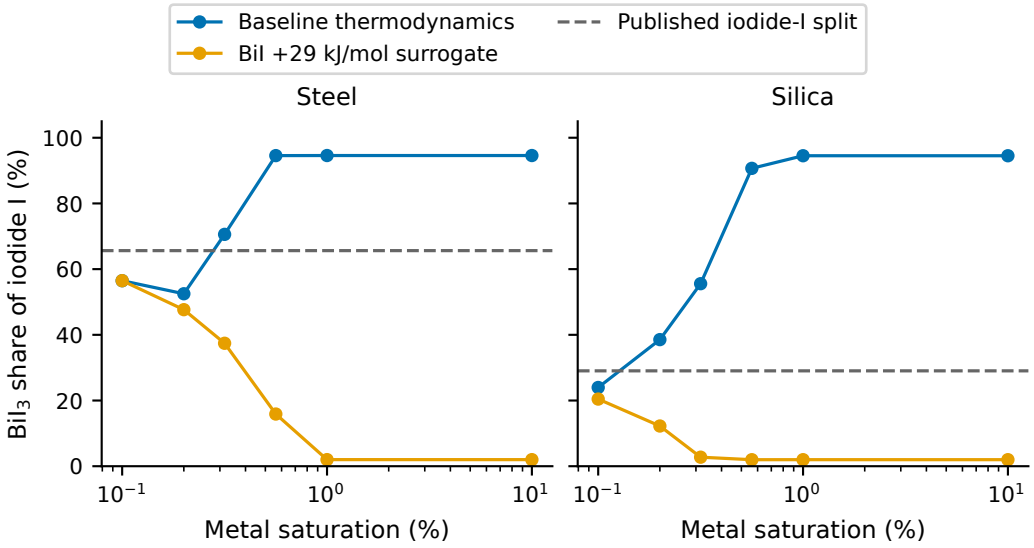



Figure 8: $BiI_3$ share of deposited iodide iodine (eq. (20)) against source loading for the baseline and the +29 kJ/mol BiI thermodynamics, for steel and silica LBE. Symbols denote completed Q2 cases; segments only connect those points. Dashed grey lines show the published $BiI_3$ share after excluding KI [1]. The +29 kJ/mol branch is a surrogate perturbation, not a second qualified thermodynamic dataset.

### 7.1 Source loading and thermodynamic input

The Q2 cases expose a sensitivity that is hidden when the iodide split is prescribed. Figure 8 shows the $BiI_3$ share of deposited iodide iodine,

$$f_{\mathrm{BiI_3}} = \frac{3N^{\mathrm{wall}}_{\mathrm{BiI_3}}}{2N^{\mathrm{wall}}_{\mathrm{PbI_2}} + 3N^{\mathrm{wall}}_{\mathrm{BiI_3}}}, \tag{20}$$

which counts only the iodine in the two modelled iodides and therefore differs from the $BiI_3$ fraction of all experimental iodine, which also contains KI.

At 1% metal saturation on the production steel mesh, the baseline thermodynamics give $f_{\mathrm{BiI_3}} = 94.58\%$ (94.57% with upwind advection). Raising the BiI(g) standard Gibbs energy by 29 kJ/mol, consistently in the formation and reactive-wall constants, gives 2.01% (2.00%). A constant shift of this size scales the formation constant by $\exp[-29{,}000/(RT)]$, approximately 0.051 at 1173 K and 0.00093 at 500 K. An independent integration of the saved species profiles of the original upwind campaign gives 94.532–94.569% and 1.922–2.004% over three meshes and the half-step run, a separation of 92.555–92.619 percentage points at every matched resolution. The response is therefore not numerical noise. What it measures is the importance of the BiI thermodynamic input for the predicted iodide partition: a single, arbitrarily sized perturbation moves the partition almost from one iodide to the other. The shift is not an estimate of the uncertainty of the BiI data, the two branches do not bound an uncertainty interval, and the comparison with the measured split is not a basis for selecting either dataset.

Source loading also matters. With baseline thermodynamics, the steel $BiI_3$ share at 0.2%, 0.316% and 0.562% metal saturation (follow-up cases, 18,432 cells, upwind) is 52.50%, 70.58% and 94.56%; the shifted branch gives 47.66%, 37.43% and 15.88%. The low-loading trend is not universally monotonic. The steel 0.1% case with baseline thermodynamics gives 56.48% and an 80.66% scan overlap (56.47% and 81.10% on the 18,432-cell upwind mesh), a useful diagnostic comparison but not an identified physical saturation. The published deposits alone cannot separate source loading, thermodynamic data and omitted chemistry; choosing a branch because it best matches one scan would be a calibration.

### 7.2 Silica temperature-coordinate inconsistency

In the silica experiment the published peak labels do not match the published temperature curve. The scan places the $PbI_2$ and $BiI_3$ maxima at 51 and 59 cm, labelled 306±24 and 127±39 °C. At those positions the digitised temperature curve gives approximately 369 and 235 °C, 63 and 108 K above the labels; on the same curve, 306 and 127 °C occur at 55.096 and 63.653 cm. The discrepancy exceeds the nominal raster temperature resolution, so the published inputs

Table 6: Silica temperature-coordinate hypotheses: computed peak positions and agreement with the fixed experimental scan for the original and two shifted temperature curves. Shifts and peak positions are in cm; raw L1 ($L_{\mathrm{raw}}$) is in percentage points and overlap in percent. Q2 uses baseline thermodynamics and 0.1% metal saturation.

| Source | $\Delta x_T$ | $x_{\mathrm{PbI_2}}$ | $x_{\mathrm{BiI_3}}$ | Raw L1 | Overlap |
|---|---|---|---|---|---|
| Q1 | 0.00 | 56 | 63 | 183.84 | 6.13 |
| Q1 | -4.10 | 51 | 59 | 40.17 | 77.80 |
| Q1 | -4.65 | 51 | 58 | 51.44 | 73.26 |
| Q2 | 0.00 | 55 | 63 | 183.84 | 6.13 |
| Q2 | -4.10 | 51 | 59 | 37.40 | 77.88 |
| Q2 | -4.65 | 51 | 58 | 47.21 | 75.54 |

Table 7: Range of the baseline-relative iodine profile change $E_{L1}$ (percent) within each completed sensitivity category. The flow-stop row compares 80 s inventories with their 60 s restart state; the other rows compare 60 s inventories. The silica temperature-coordinate row uses the production cases (41,472 cells, limited-linear advection); all other rows are upwind follow-up cases on 18,432 cells compared with their own upwind baselines; table 8 repeats the diffusivity and accommodation rows with the production scheme for two cases. Case counts and variations are listed in Table S1 of the supplementary material.

| Assumption | Minimum | Maximum |
|---|---|---|
| Gas diffusion | 0.15 | 14.20 |
| Iodide accommodation | 0.25 | 4.21 |
| Source position | 0.08 | 0.19 |
| Q2 metal loading | 0.00 | 112.14 |
| Flow reference | 5.09 | 10.93 |
| Silica temperature coordinates | 187.10 | 193.96 |
| Flow-stop relaxation | 1.10 | 4.63 |

contain a temperature-coordinate inconsistency; they do not say which measurement, annotation or registration is in error.

Two diagnostic mappings shift only the silica temperature curve upstream, by 4.096 or 4.653 cm, so that one of these curve crossings aligns with the corresponding labelled position: $T_\delta(x) = T_0(x + \delta)$ for $\delta > 0$, with $\Delta x_T = -\delta$. The iodine scan stays fixed. The shifts are derived from the published curve and labels, not optimised against the CFD profiles.

Table 6 and fig. 9 show that the shifts move the Q1 maxima from 56/63 cm to 51/59 or 51/58 cm and raise the normalised overlap from 6.13% to 77.80% or 73.26%. The low-loading Q2 variants place the maxima at 51/59 or 51/58 cm, with overlaps of 77.88% and 75.54%. On the 18,432-cell upwind mesh the order of the two shifts was the opposite (75.29% and 81.00% for Q1; 79.16% and 83.33% for Q2), so the overlap does not rank the two hypotheses robustly. The two shifts also give different peak temperatures, and no single mapping reconciles both labels. The shifted low-loading Q2 cases keep a $BiI_3$ share near 23.97%, about five percentage points below the published iodide-only split, so correct peak positions do not ensure the correct partition.

These results demonstrate the inconsistency and show that a plausible coordinate change has a larger effect on the computed agreement than the tested source-position or time-step changes. They do not resolve the inconsistency: both shifts use labels from the same experiment, neither is confirmed, and the improved agreement is conditional on an unconfirmed hypothesis. Resolving it requires the original thermocouple profile and its coordinate registration.

### 7.3 Transport, kinetics, flow reference and flow stop

Table 7 reports, for each tested category, the range of iodine profile changes $E_{L1}$ (eq. (16)) from the designated baseline. These are responses to finite hypothetical changes, not statistical confidence limits. The source-loading row uses its 1% reference case. The temperature-coordinate row measures changes from the original silica input and can approach 200% when the profile supports move apart.

Diffusion alternatives change profiles by up to 14.20%. A 273.15 K flow reference raises the inferred helium mass flow by 9.15% relative to 298.15 K and changes profiles by up to 10.93%. Iodide accommodation changes give up to 4.21%, whereas moving the prescribed source region by $\pm 1.5$ cm gives at most 0.19%, a statement about the empty-tube source model that cannot rule out an effect of a real boat on the flow. Fig. S1 of the supplementary

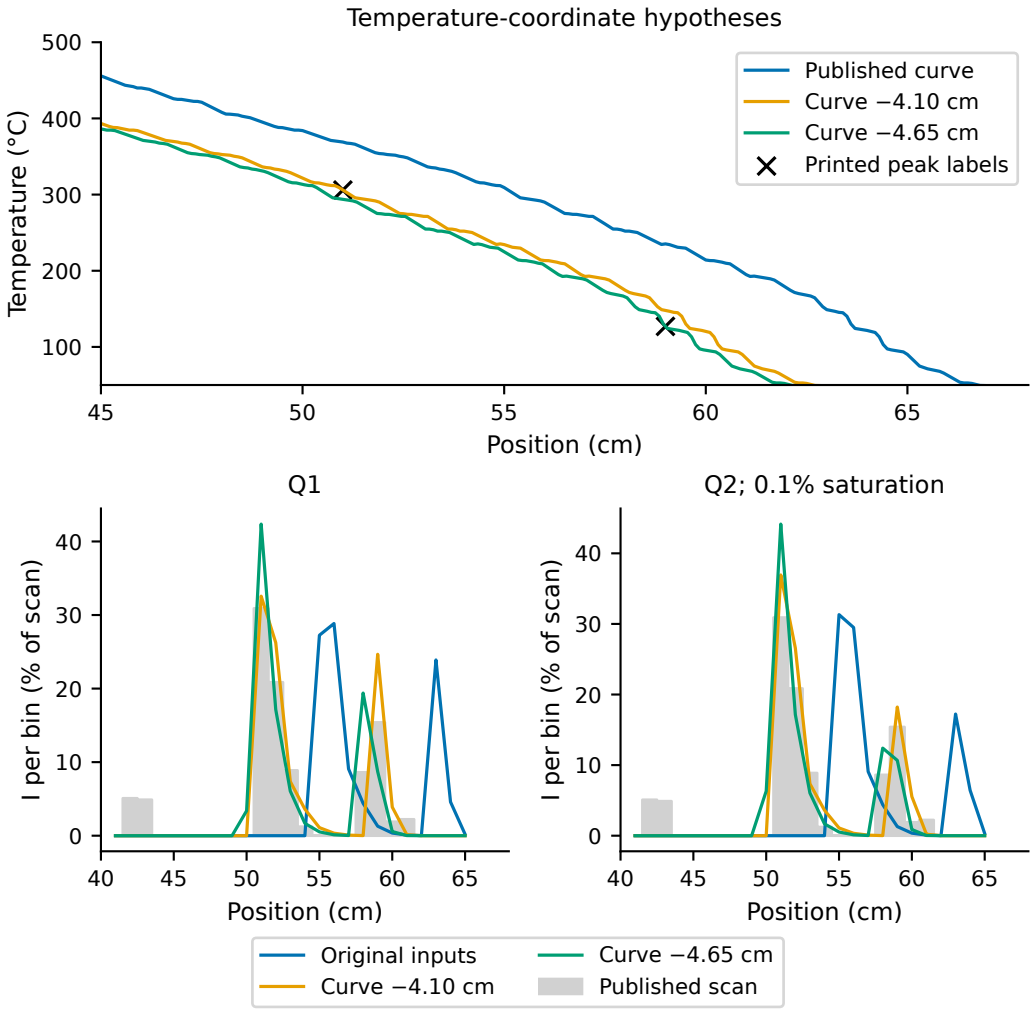



Figure 9: Input-derived silica mappings and their consequences. Top: the original reconstructed temperature curve, the two shifts and the printed peak labels. Bottom: common-scan-normalised iodine profiles for Q1 and low-loading Q2. The improved agreement is conditional on an unconfirmed coordinate hypothesis. Published data are adapted from Liu et al. [1].

material shows the maxima. They are differences between upwind cases on the 18,432-cell mesh, which itself differs from the production scheme by 12–14% for the two cases of the scheme study (section 5). For the diffusivity and accommodation variations, table 8 shows how much of this survives the scheme change. With the production scheme, $\alpha = 0.1$ changes the iodine profile by 0.15–0.49% and $\alpha = 0.01$ by 1.90–5.14%, against 1.07–16.65% for halving or doubling $D$; the pure $PbI_2$ responses are slightly larger than with upwind and nearly symmetric in $D$. The steel response to $D$ is not: halving $D$ changes the iodine profile by only 1.07% (upwind 4.61%), because the $BiI_3$ profile, which carries most of the iodine, barely moves (0.60% against 4.41%), whereas doubling $D$ gives 12.53% (upwind 6.68%). The diffusivity response is therefore case-dependent and not linear in $D$, and in the steel case $\alpha = 0.01$ changes the profile more than halving $D$. The other categories of table 7 have upwind differences only, so effects of comparable size there are not ranked precisely. In both production-scheme cases, $\alpha = 0.1$ changes the profile far less than either diffusivity change. Together with the tube-scale Damköhler numbers of section 4.2 and the channel sweep of fig. 3, which places $\alpha = 1$ and 0.1 on the diffusion-limited plateau and $\alpha = 0.01$ at its edge, this indicates that radial diffusion, not interfacial kinetics, limits the wall exchange in these columns; the wall temperature acts mainly through the equilibrium pressure.

The four fixed-temperature flow-stop cases change the deposited iodine profiles by 1.10–4.63% over 20 s, with centroid shifts of at most 0.0422 cm. Residual gas can therefore alter wall inventories after release stops, but because these runs neither cool the column nor flush it, they do not predict the end-of-experiment redistribution.

Table 8: Diffusivity and accommodation variations with the production scheme (LL: limited-linear, 41,472 cells) and with upwind on 18,432 cells, each relative to its own baseline: iodine profile change $E_{L1}$ (percent) and centroid shift $\Delta x_c$ (cm). Steel Q2 uses baseline thermodynamics. Total deposits change by at most 0.02% and the $BiI_3$ share of iodide iodine by at most 0.03 percentage points.

| Case | Variation | $E_{L1}$ LL | $E_{L1}$ upwind | $\Delta x_c$ LL | $\Delta x_c$ upwind |
|---|---|---|---|---|---|
| Pure $PbI_2$ | $D/2$ | 16.65 | 14.20 | +0.230 | +0.230 |
| | $2D$ | 8.98 | 7.25 | −0.085 | −0.088 |
| | $\alpha = 0.1$ | 0.49 | 0.41 | +0.005 | +0.005 |
| | $\alpha = 0.01$ | 5.14 | 4.21 | +0.056 | +0.057 |
| Steel Q2, 1% | $D/2$ | 1.07 | 4.61 | +0.003 | +0.032 |
| | $2D$ | 12.53 | 6.68 | +0.069 | +0.036 |
| | $\alpha = 0.1$ | 0.15 | 0.36 | +0.001 | +0.002 |
| | $\alpha = 0.01$ | 1.90 | 3.67 | +0.012 | +0.024 |

# 8 Discussion: what the evidence supports

## 8.1 Conservation, resolution and experimental agreement

The results separate four requirements that a deposition calculation must satisfy. Element balance tests the consistency of the discrete transport–equilibrium–wall coupling; the Graetz–Robin test verifies the wall discretisation against a reference solution; resolution studies test how much the computed observables depend on numerical choices; and experimental comparison tests the combined consequences of the numerical model, its physical assumptions and its input data. Passing the first two does not establish the others.

Here the raw material residues are at round-off and the half-cell wall law converges at second order, yet local profile shapes remain resolution-sensitive while integrals are stable, and nearly equal bins can shift a reported peak temperature by tens of kelvin. Profile differences, centroids and deposited amounts should therefore accompany peak comparisons. The physical-input responses are finite responses over selected cases; they cannot be added in quadrature or read as a universal parameter ranking, and several are comparable in magnitude to the unresolved spatial differences.

## 8.2 What the experimental comparisons can identify

Q1 uses deposit-derived compound masses for LBE, so agreement in the iodide split partly reflects information supplied to the calculation. Q2 lets the split emerge but combines a deposit-derived iodine loading with an unmeasured metal saturation; the computed split responds strongly to both the BiI thermodynamic input and the source loading, so one observed split constrains several coupled assumptions rather than identifying one of them. The silica scan and its printed temperatures are mutually inconsistent, and the two diagnostic shifts expose rather than resolve this. The untuned Lobresco comparison gives onset temperatures closer to the measurements than both published models, but with different carrier, vapour-pressure and source assumptions, a rule-dependent T5_60 value and profiles displaced upstream; it is encouraging for the transport and wall law, not a demonstration of them. The practical contribution is therefore a framework for testing explicit input hypotheses while retaining conservation and exposing profile discrepancies. Fitting an interfacial coefficient to a single peak under these conditions could compensate for an error elsewhere in the model.

## 8.3 Limitations

The limitations are collected here and referred to elsewhere.

- *Physics not represented:* a frozen, trace-loaded carrier with the wall temperature imposed over each cross-section, so no buoyancy-driven secondary flow (possibly significant locally in a horizontal column; orientation unknown), no conjugate heat transfer and no non-dilute effects near a source; no aerosols or nucleation (although the computed core is supersaturated, fig. 4), no Soret transport, KI or water/oxygen chemistry, substrate reactions or boat obstruction; instantaneous ideal-gas equilibrium and an irreversible BiI wall channel in place of unmeasured kinetics.
- *Inputs:* digitised rather than tabulated wall temperatures; an unconfirmed coordinate origin and the silica inconsistency; assumed controller reference conditions and column pressure; deposit-conditioned source masses and an unmeasured metal saturation; BiI and constructed $BiI_3$(l) thermodynamic records that need independent qualification.

- *Protocol:* 60 s source-driven runs at the three-hour mean release rate, flow-stop runs at fixed temperature, and no simulation of the cooling or withdrawal at the end of any experiment.
- *Numerics:* bounded-scheme refinement to 93,312 cells beyond one case (with upwind, four cases change by 2.13–7.30% from 41,472 to 93,312 cells), a consistent observed order for the iodine profiles, production-scheme repeats of the follow-up categories other than the diffusivity and accommodation variations of two baselines, a reference-solution verification of the HKS law with tabulated equilibrium pressures (the channel sweep checks only its limits) are outstanding.

Each item addresses an ambiguity demonstrated by the completed cases. The most direct experimental constraints would be the original thermocouple and scan coordinates, controller reference conditions, an independently measured release history and the column orientation and pressure. None of the omitted effects can be inferred from the present sensitivity ranges.

# 9 Conclusions

We have applied a conservative finite-volume treatment of nine Pb–Bi–I gases and four shared condensed wall inventories, with element-conserving gas equilibrium and reversible face-flux wall exchange, to 15 baseline and 64 follow-up cases, 19 production re-runs and a scheme, mesh and iteration study. The columns are laminar helium flows ($\mathrm{Re} \approx 0.7$–3) in tubes with a steep axial wall-temperature gradient, where axial advection, radial diffusion and the interfacial exchange law together set the deposition profile. The worst raw element residue is $8.85 \times 10^{-13}$ of supplied material in the follow-up cases and $1.41 \times 10^{-12}$ in the production cases; conservation is a numerical property of the model, not evidence that all physical pathways are represented. On a Graetz–Robin problem the half-cell wall law converges at second order to the reference decay rate, whereas applying the kinetic law to the cell-centre value is first order; this verifies the wall discretisation, not the convergence of the iodine calculations. Within the model, radial diffusion rather than interfacial kinetics limits the wall exchange. A sweep of the wall Damköhler number over six decades on a plane channel reproduces the kinetic limit, the diffusion limit and the series-resistance transition between them, and places the production accommodation coefficients on the diffusion-limited plateau or, for $\alpha = 0.01$, at its edge; the deposition onset does not move across the sweep, because saturation fixes it.

Integral results are much less sensitive to resolution than local profiles. Halving the time step changes iodine profiles by at most 0.323%, and eight instead of three species correctors by at most 0.021%. The spatial error is larger and is dominated by the numerical diffusion of first-order upwind advection, which differs from the limited-linear scheme by 7.7–20.8% at equal mesh in the steel case of the scheme study, whereas two bounded schemes differ by at most 2.8%. With limited-linear advection that case changes by 1.98% between 41,472 and 93,312 cells, but the apparent orders are inconsistent, so we quote this change rather than a formal grid convergence index. With upwind advection the same refinement changes the four checked cases by 2.13–7.30%, and three of them more than that steel case, so its bounded-scheme change is not a bound for the other production profiles. Closely competing 1 cm bins can shift a reported peak temperature by tens of kelvin, so peaks are reported with their smoothed maximum and centroid temperature as well. Repeated with the production scheme for two baselines, $\alpha = 0.1$ changes the iodine profiles by at most 0.49% and halving or doubling the diffusivity by 1.07–16.65%; the steel response to the diffusivity is strongly asymmetric, so it is case-dependent and not linear.

The computed iodide partition depends strongly on thermodynamic input. In the refined steel case, a $+29$ kJ/mol shift of the BiI gas Gibbs energy changes the $BiI_3$ iodine share from 94.58% to 2.01%, and a comparable separation persists at every tested resolution. This shows how much the prediction depends on the BiI data; it is neither an uncertainty interval nor a criterion for choosing a dataset. For silica, temperature-coordinate shifts derived from the published curve and labels reduce 4–5 cm position discrepancies to 0–1 cm. This demonstrates an inconsistency in the published coordinates without resolving it; the overlap ranking of the two shifts even reverses between advection schemes.

On five single-species $PbI_2$ experiments of Lobresco et al., the untuned model gives deposition-onset temperatures with an RMS deviation of 18 K, against 35 and 35 K for the two models of that study. The result is promising but conditional: it depends on applying the onset rule to the main deposit (from the inlet, the RMS would be 135 K), the near-source mole fraction of 0.33–0.36 violates the dilute frozen-carrier assumption, and the release history is unconstrained below 6 s. Because carrier, vapour-pressure and source assumptions also differ from the published models, the improvement cannot yet be attributed to the wall treatment, and the computed peaks lie 1–4 cm upstream with summed profile differences of 0.37–0.89 of the sample amount.

The evidence supports a conservative method with a verified wall discretisation and an explicit assessment of uncertain inputs, not independent predictive validation. The requirements for the latter are listed in section 8.3.

## CRediT authorship contribution statement

**Jan Kren**: [AUTHOR TO CONFIRM: CRediT roles]. **Bojan Ničeno**: [AUTHOR TO CONFIRM: CRediT roles].

## Declaration of competing interest

[AUTHOR TO CONFIRM: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

## Declaration of generative AI and AI-assisted technologies in the writing process

[AUTHOR TO CONFIRM: whether such tools were used and, if so, Elsevier's statement: "During the preparation of this work the authors used [tool] in order to [reason]. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication." Remove this section if no such tool was used.]

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## Data availability

The solver, the benchmark preparation scripts and the data that support the findings of this study are openly available in the LESTO repository at https://github.com/JanKren/LESTO, [AUTHOR TO CONFIRM: archived release or DOI, and that the branch holding these files is public]. These files comprise the completed baseline, follow-up, production, scheme and mesh, and iteration-check metrics, the Graetz–Robin verification records, saved axial profiles, generated tables, figure-generation code and SHA-256 provenance records. The experimental curves were digitised from Liu et al. [1]; they reconstruct published figures, are not author-supplied measurements, and are adapted under the source paper's CC BY 4.0 license. The measured profiles, wall temperatures and model curves of Lobresco et al. [6] were digitised in the same way and are adapted under that paper's CC BY 4.0 license; the case set-up and final profiles of our comparison runs are included. Some thermodynamic coefficient files are external inputs that are not redistributed; their local identifiers and the provenance of the derived tables are recorded with the benchmark. Reproducing the CFD calculations from the coefficients requires access to those inputs, whereas regenerating the figures from the saved results does not.

## Supplementary material

Supplementary material associated with this article (case inventory, detailed material-ledger and solver diagnostics, scalar mesh checks of the upwind campaign, comparison-bin-width analysis, property and thermodynamic records, and the complete Graetz–Robin verification table) is provided as a separate file.

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